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LRSM WR→tb Channel Analysis

Decay channel: WR→e/μ + N→e/μ + WR*→tb

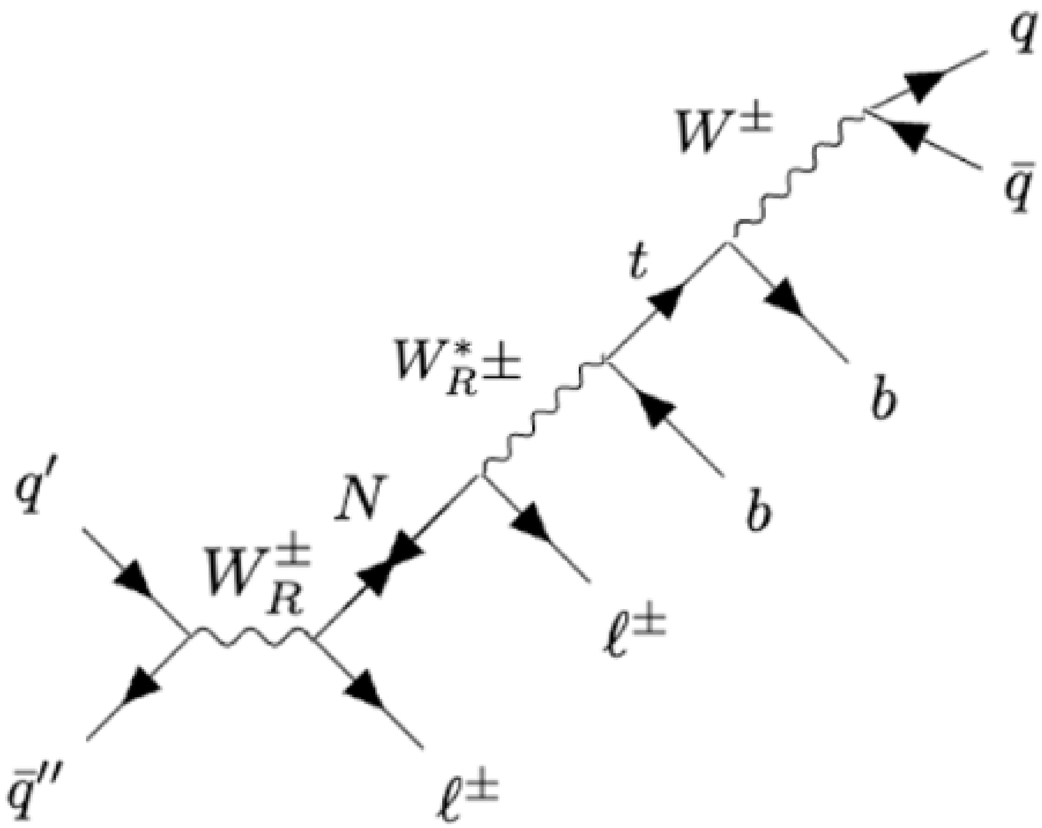

Figure 1: Figure 1: Channel diagram

1. Introduction

1.1 Motivation and Channel Selection

Previous research focused on the **WR high, N low** region where N particles are highly boosted, causing jets to cluster together. The primary decay modes studied were WR*→ud, sc channels, which produce at most 3 substructures (q, q, τ) within a single AK8 jet.

In this channel, due to using 2 lepton and top tagging and b tagging also with high m_ll, we can reduce lot of backgrounds and also improvements in WR ~ N region in the previous study.

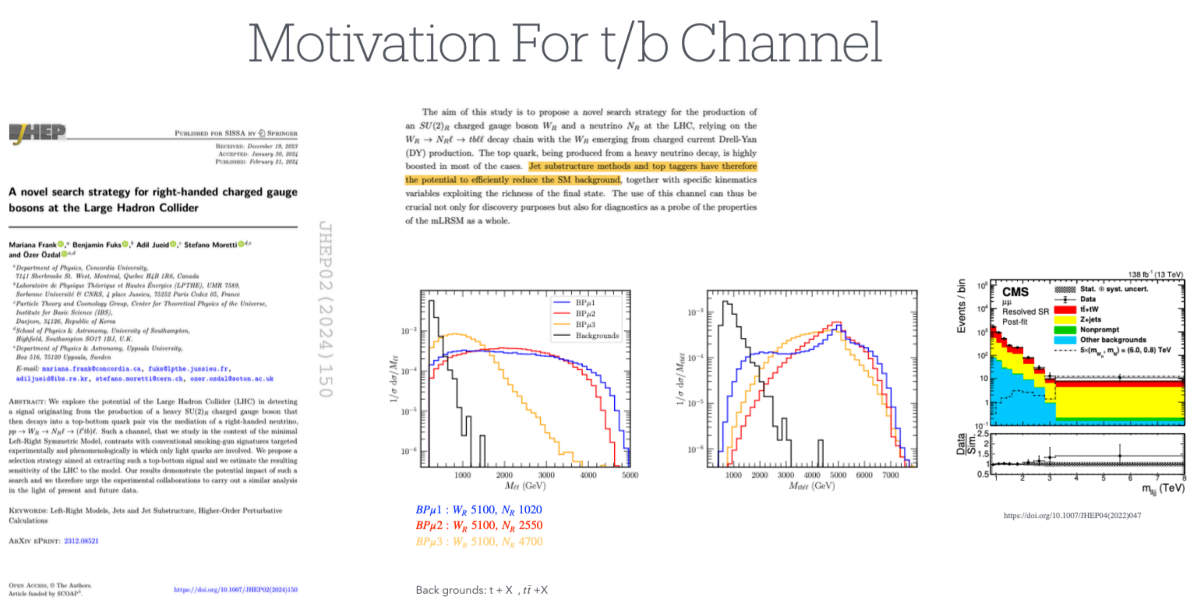

Figure 2: Figure 2: Previous study motivation

Previous study data (EXO-20-002)

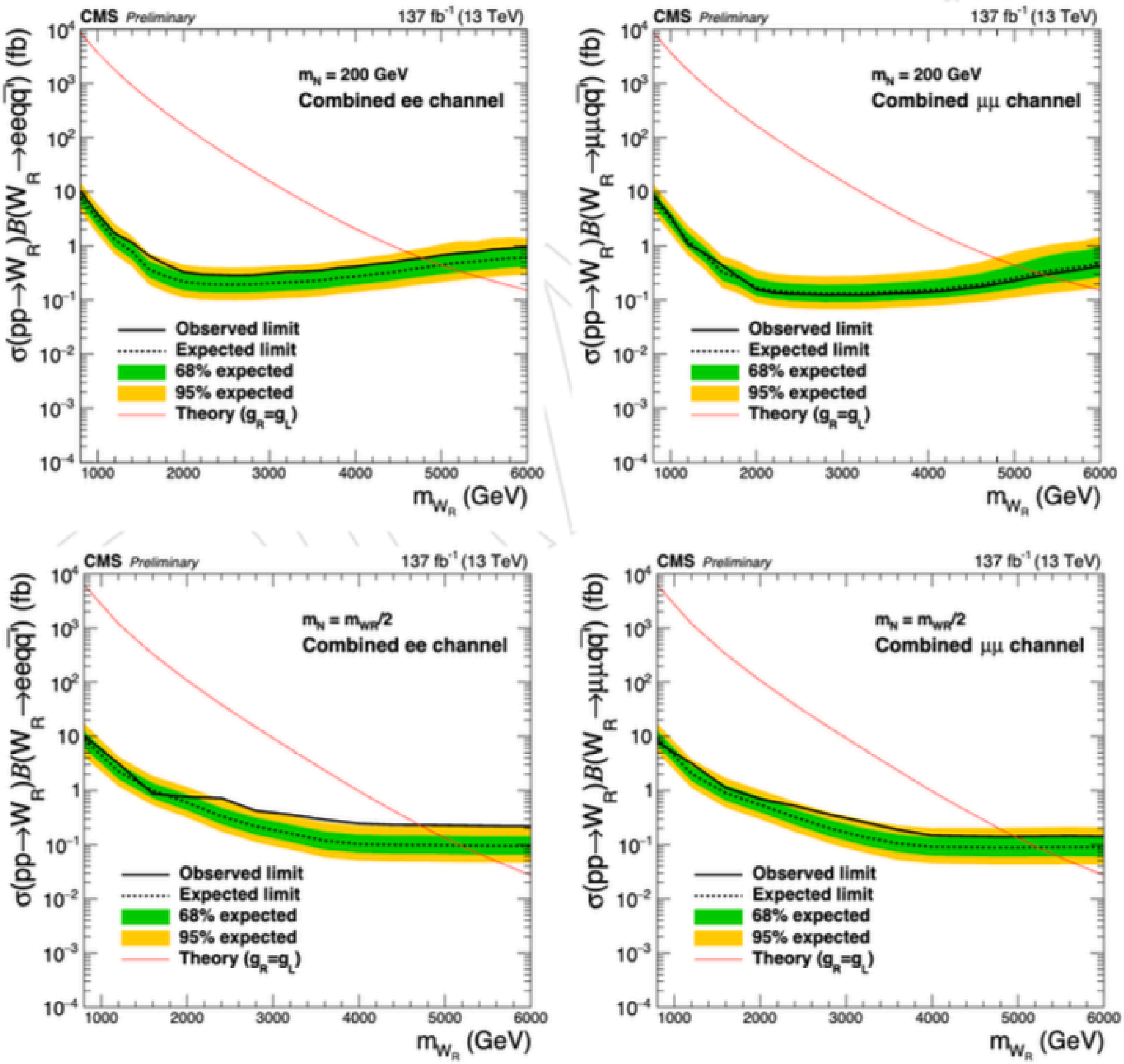

Figure 3: Figure 3: Cross-section comparison

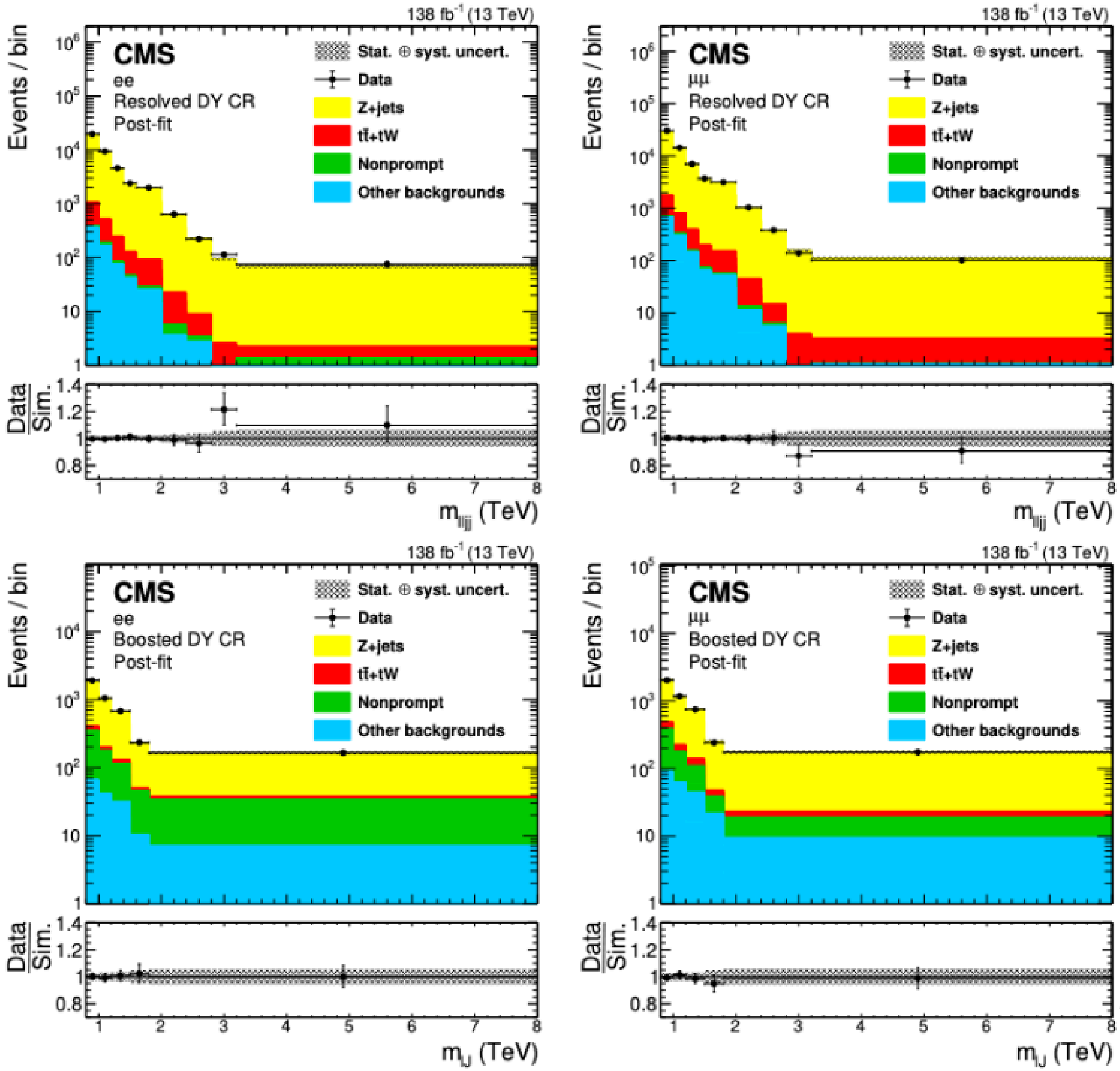

Figure 4: Figure 4: Mass limits from previous analysis

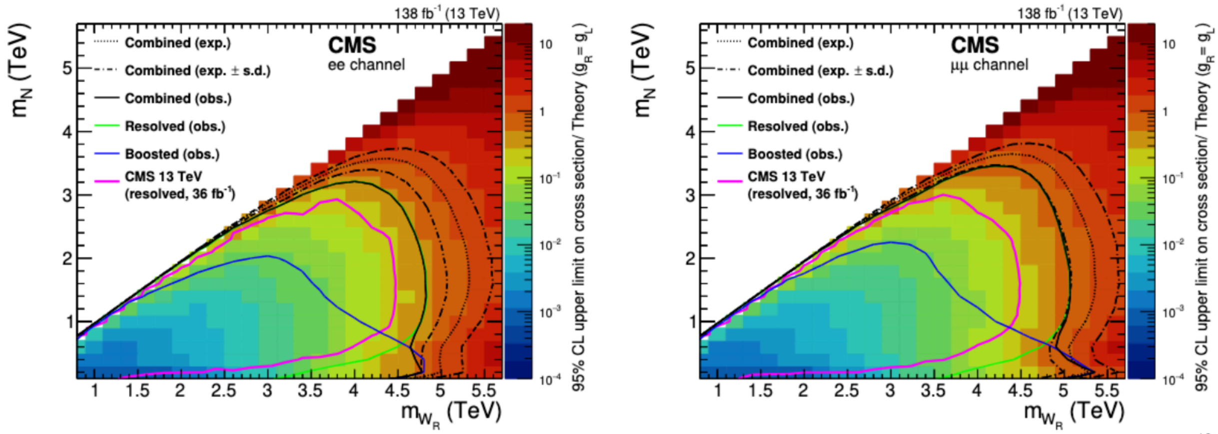

Figure 5: Figure 5: Exclusion contours

	Electron channel		Muon channel	
$]M_{\ell\ell jj}^{\min}, M_{\ell\ell jj}^{\max}]$	Data	Background	Data	Background
$]800, 1000]$ GeV	1106.0	1103.5 ± 26.607	1639.0	1670.7 ± 39.774
$]1000, 1200]$ GeV	646.0	631.51 ± 16.968	946.0	925.99 ± 23.917
$]1200, 1400]$ GeV	332.0	323.23 ± 10.736	518.0	500.33 ± 14.869
$]1400, 1600]$ GeV	170.0	169.69 ± 6.8418	268.0	263.88 ± 9.3498
$]1600, 2000]$ GeV	143.0	157.55 ± 9.505	216.0	215.18 ± 8.2146
$]2000, 2400]$ GeV	62.0	52.327 ± 3.9676	80.0	73.482 ± 4.4654
$]2400, 2800]$ GeV	25.0	19.567 ± 1.5493	30.0	25.943 ± 2.3125
$]2800, 3200]$ GeV	10.0	8.9907 ± 1.209	13.0	9.7557 ± 1.1603
$]3200, 8000]$ GeV	13.0	6.2463 ± 0.77892	11.0	7.8119 ± 0.84286

arXiv:2312.08521v2

EXO 20 -002 data / bkg rates
(light quark channel)

Figure 6: Figure 6: Additional results

1.2 Challenge with tb Channel

Extending the analysis to include the **tb decay channel** presents kinematic challenges:

- **Top quark rapid decay**: t → b + W, where W further decays to either:
  - Hadronic: qq̄ (~70% BR)
  - Leptonic: l_nu (~30% BR)

- **Boosted regime complexity**: In the WR » N mass regime, all decay products (3 jets from top, b quark, lepton) become highly boosted, creating 5 substructures within a single cone - making reconstruction difficult.

1.3 Proposed Solution: WR ~ N Mass Region

Strategy: Target the **WR ~ N** mass region where:

1. **N becomes resolved**: Lepton and off-shell WR* separate
2. **Similar kinematics**: Off-shell WR* has similar mass to N, leading to comparable momentum
3. **Separable final states**: When WR* → tb, the top and bottom quarks can be distinguished using AK4 jets
4. **T,B-tagging capability**: b quarks (from and WR* decay) can be identified using b-tagging, and top quark can be Top tagged

1.4 Kinematic Regimes

High pT scenario (high mass WR, low mass N): - 3 objects contained within single cone - Boosted top jet reconstruction

Low pT scenario (WR ~ N mass region): - WR* cone splits into top and b cone - Resolved reconstruction possible

Reference masses: - Top quark: ~173 GeV - Bottom quark: ~4 GeV - W boson: ~80 GeV (provides moderate boost to decay products)

1.5 Analysis Variables and Cuts

Key invariant masses: - M_{ll→tb}: Full system invariant mass - M_{ll}: Dilepton invariant mass (peaks at ~90 GeV for Z resonance)

Selection considerations: - Fat jet pT cut at 200 GeV may handle lepton selection - In boosted regimes, leptons correlate with jet boost → additional lepton cuts may be needed - M_ll particularly useful for boosted leptons due to Z peak structure - More study needed

1.6 Top Decay Complications in LRSM

Chirality constraints: Since q_T originates from W_R* with right-handed chirality, all subsequent decay products maintain right-handed chirality. This leads to t → W_R b_R instead of the SM decay t → W_L b.

Key challenges:

1. **Non-SM decay mode**: Need to account for W_R b_R decay channel separately
2. **Off-shell WR suppression**: Since W_R decays off-shell near top mass, cross-section significantly reduced for massive W_R. Leptonic channels become negligible.
3. **Chirality flip probability**: For WL production, chirality flip required with probability ~ m_T/E_T
   - Example: 3 TeV WR → t(1.5 TeV) + b(1.5 TeV) → <10% flip probability
   - Results in SM-like W boson production

Mass regime dependencies:

- **Low WR (top-resolved)**: Both WL and WR decay modes possible with finite probabilities
- **High WR (top-boosted)**: Predominantly WR decay, difficult to observe due to off-shell nature

Alternative approach: In high WR, low N region, if initial WR is off-shell, reduced top boost enables WL decay mode.

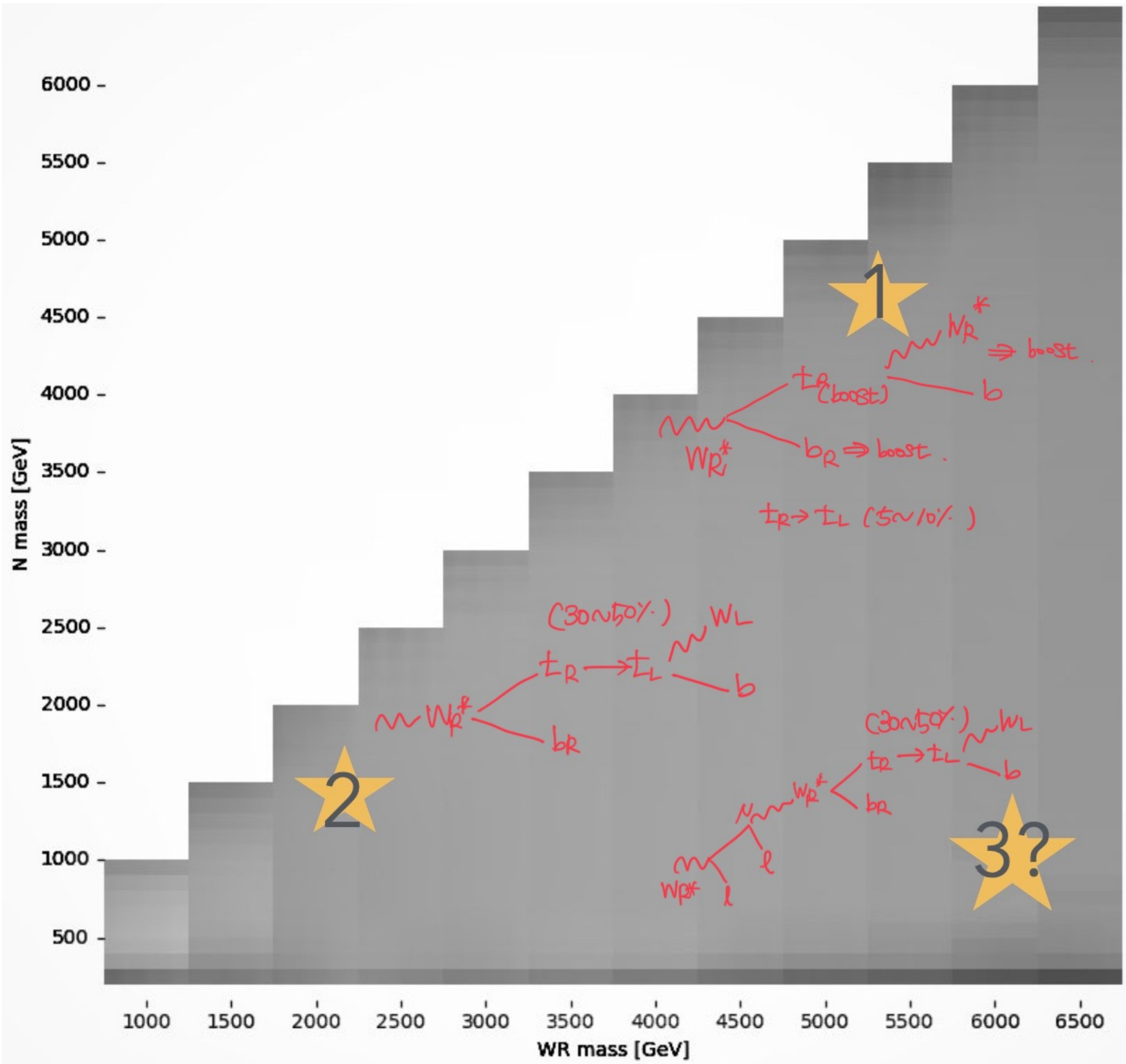

Figure 7: Figure 7: Kinematic diagram

MadGraph simulation setup: - Top couples to both WR and WL: t → (WR + WL)b vs t → WL b - Off-shell WR → qq: t → (qq)b process - Comparative study: (WL + WR) vs (WL) only - Investigation of automatic chirality flip handling in LRSM

Results: - Combined (WL + WR) shows ~1% higher cross-section than WL only - **Conclusion**: WL production dominates across most parameter space (off-shell WR production less favorable than chirality flip) - Including tb channel increases cross-section by ~10% compared to standard approach

Decay width comparison (3 TeV WR): - `generate t > q q b / w+`: Width = 1.1 × 10^{-8} ± 1.4 × 10^{-11} GeV - `generate t > q q b / wr+`: Width = 0.97 ± 0.0007 GeV - **Result**: WR decay channel negligible at high masses

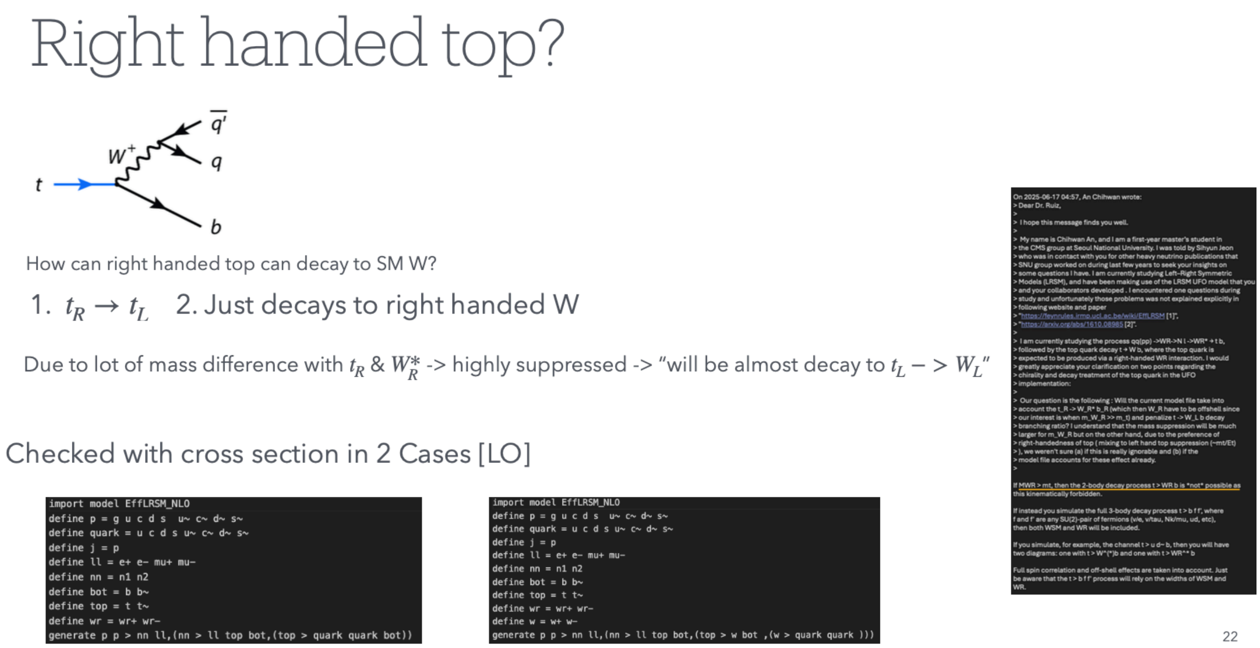

Figure 8: Figure 8: Cross-section comparison WL vs WR

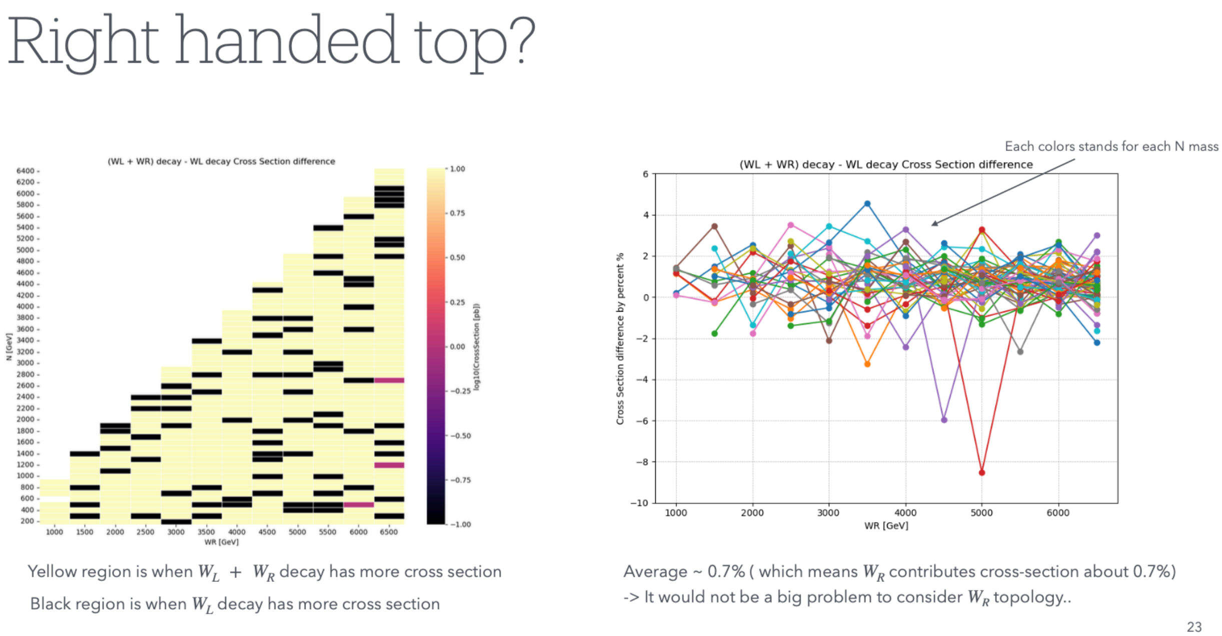

Figure 9: Figure 9: Cross-section analysis

Checked with WL + WR both decay mode and the case with decays with only SM W. There was no significance pattern when does WR contributes to cross section.

Yellow region means when WL + WR decays it has more cross section and it was order of less than 5% in LO. Also black region is when WL + WR makes less cross section and DONT KNOW WHY yet why this happens, also in LO.

1.7 Cross-Section Studies

Cross section is checked with LO gridpacks.

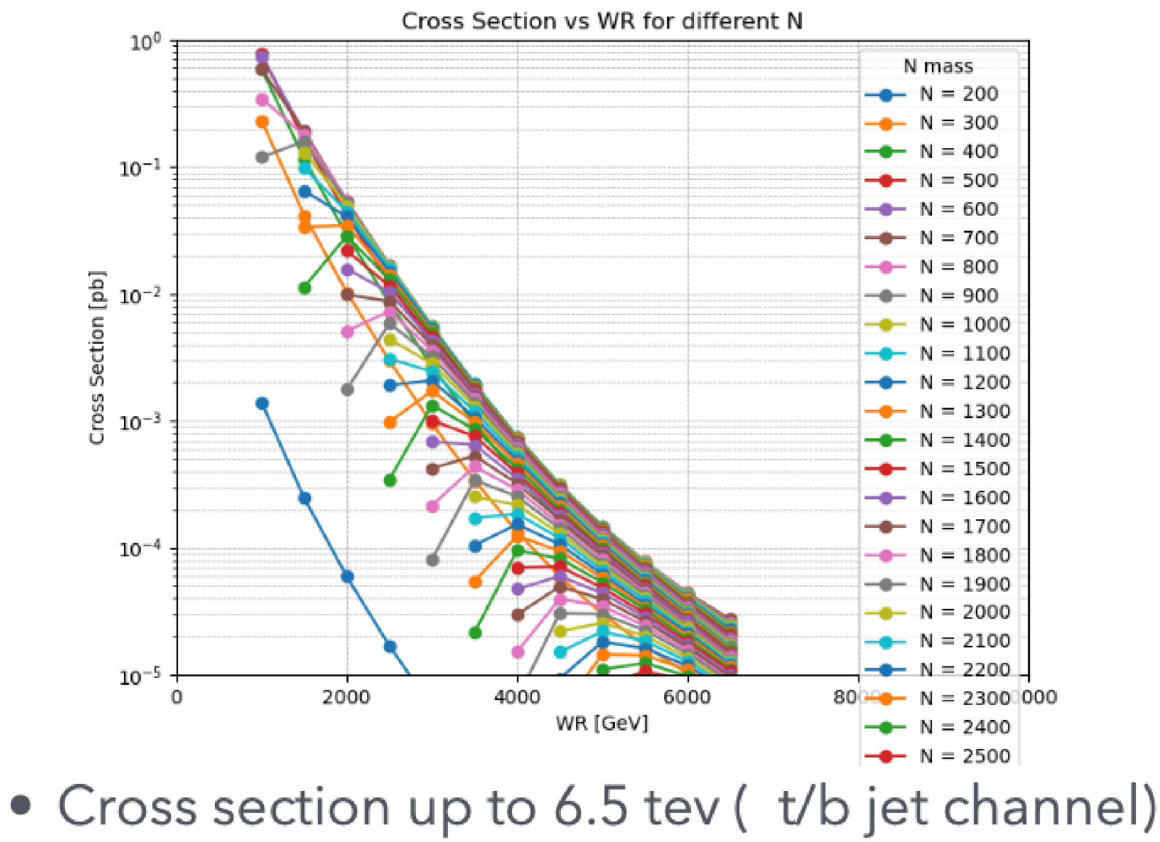

Figure 10: Figure 10: Cross-section gridpack results

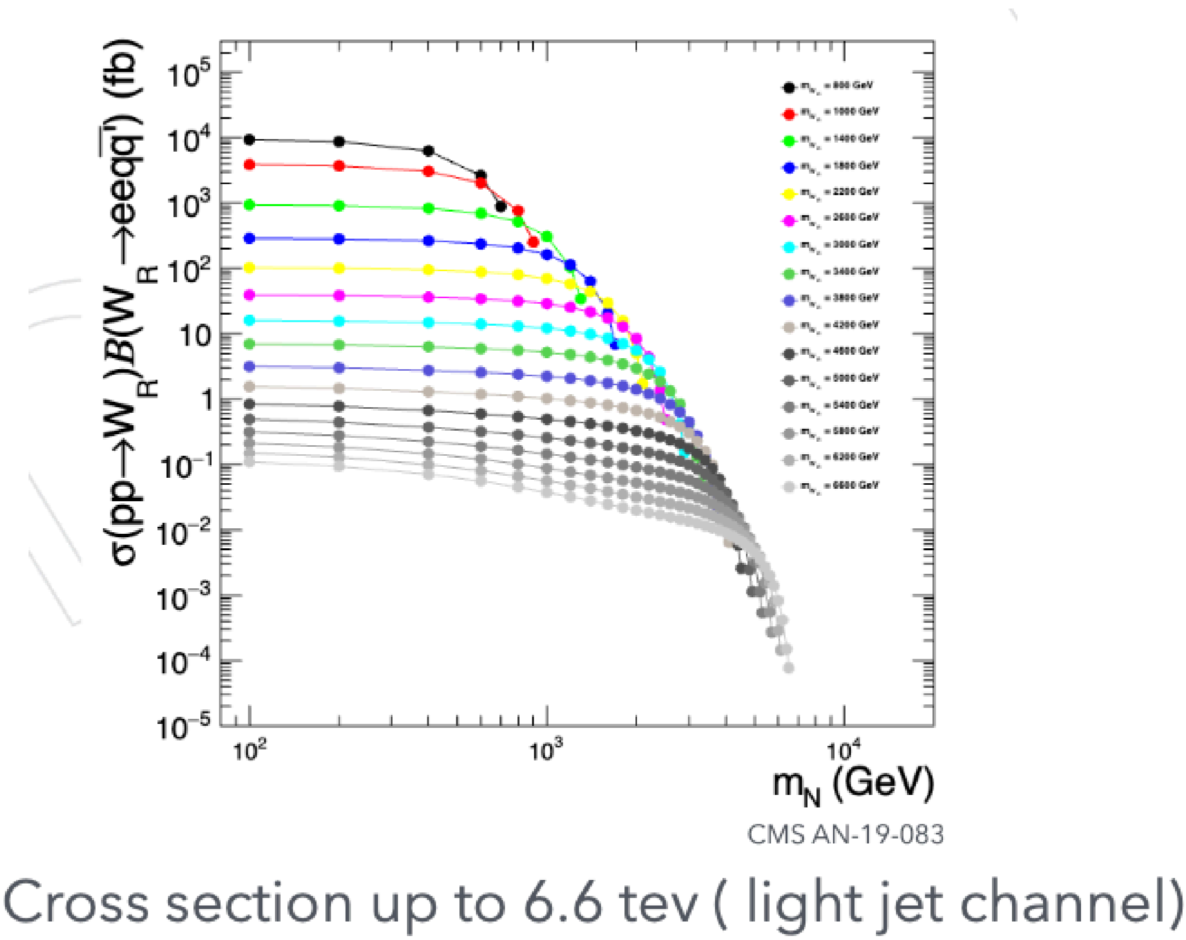

Figure 11: Figure 11: Cross sections for exo 20-002

Cross sections for exo 20-002

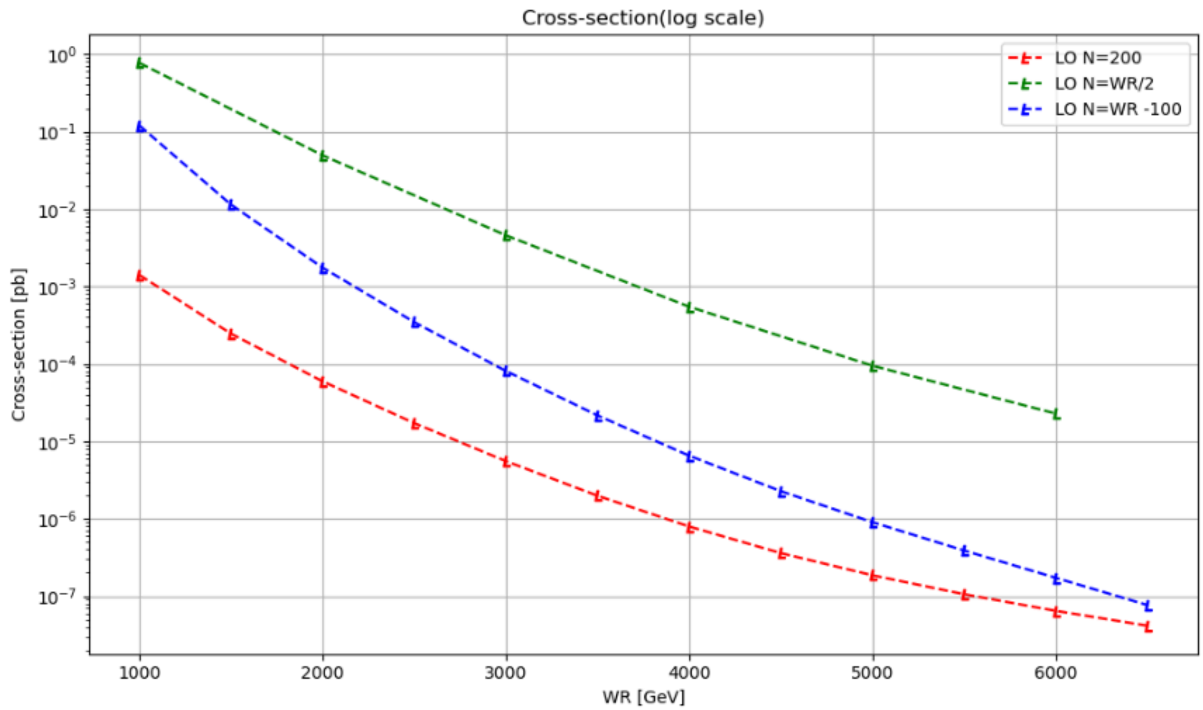

Figure 12: Figure 12: Cross-section detailed analysis

Cross section when N is ~ WR/2 has the most cross section, because N = 200 it is constraint due to top mass, and also N ~ WR has no much kinematic phase space to decay.

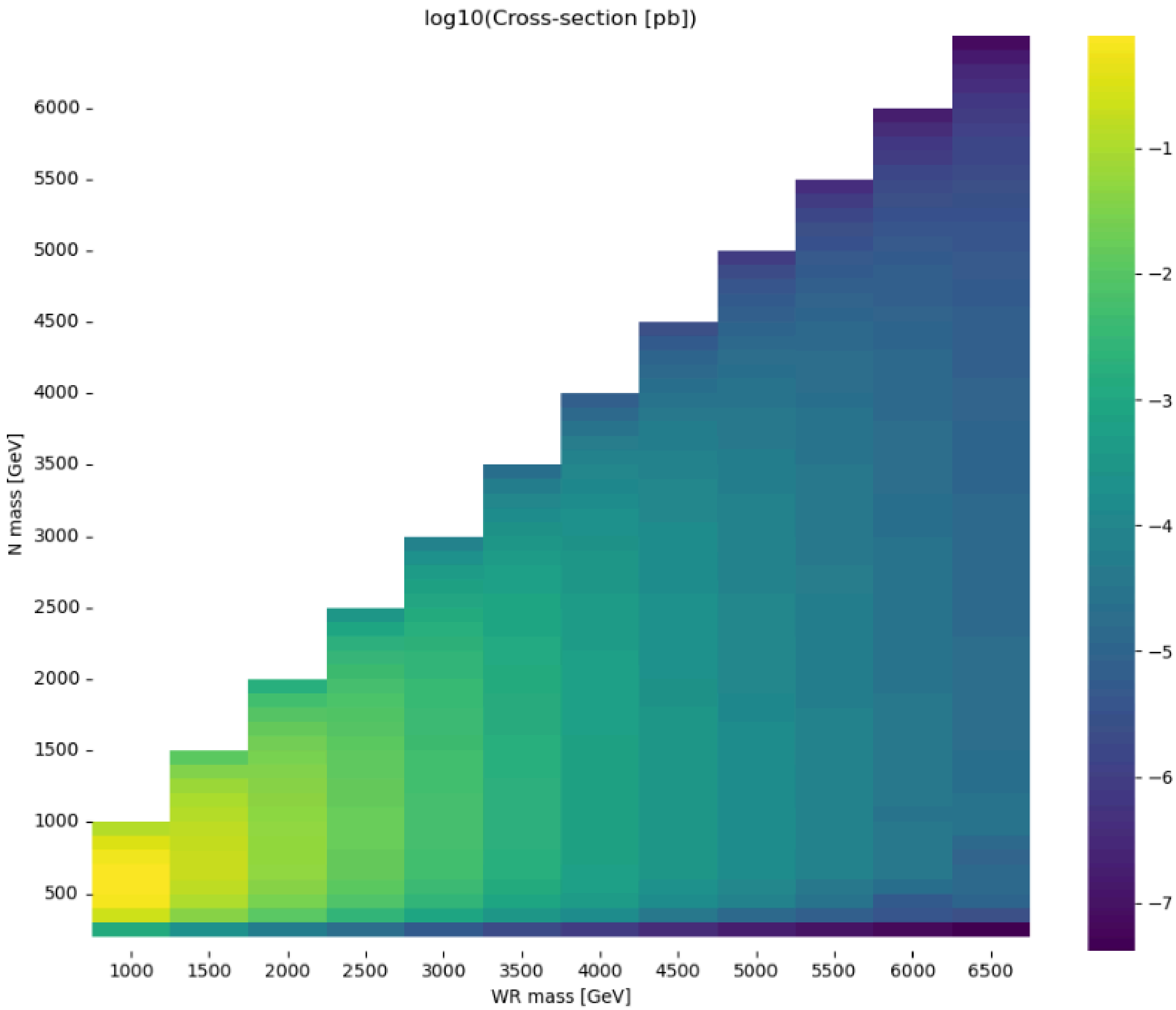

Figure 13: Figure 13: Full cross section

Full cross section

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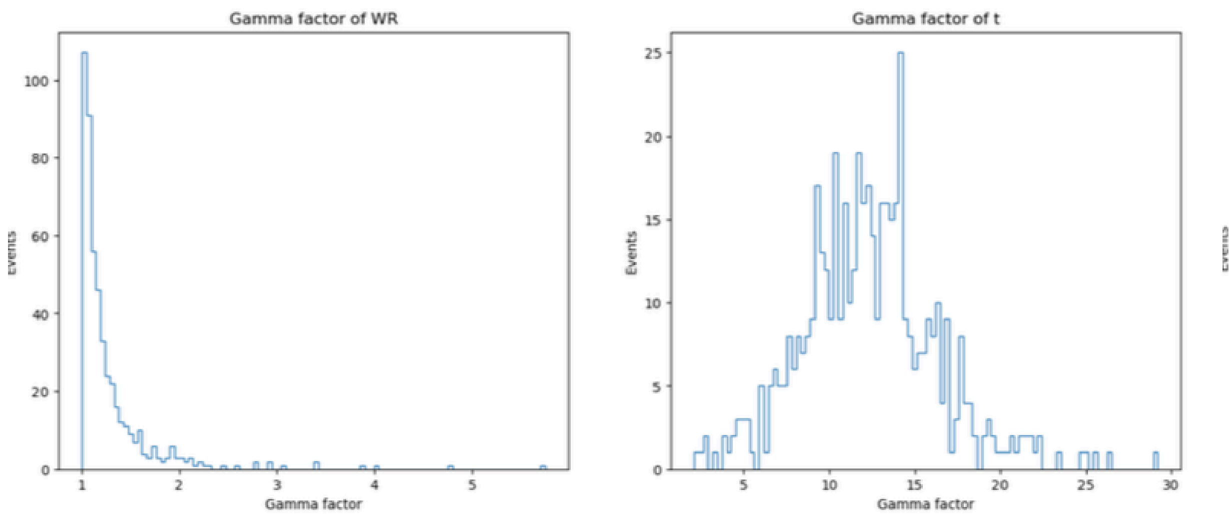
2. Signal Analysis

2.1 Kinematics and Event Topology

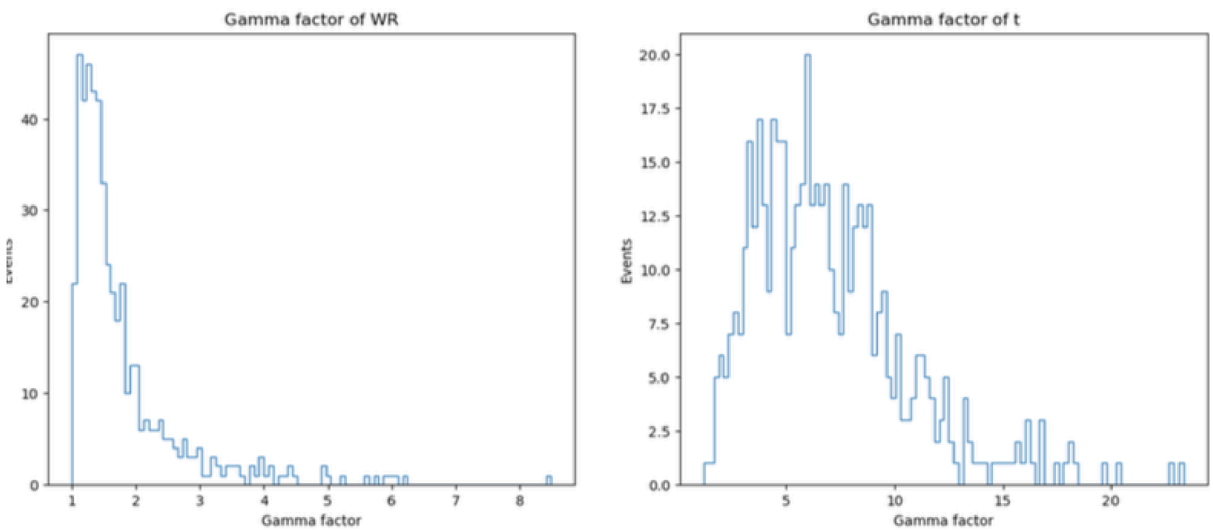
Analysis approach: Event kinematics studied using LHE files (parton level) to understand ΔR and pT distributions.

Boost Factors for non observable objects (WR, top)



W_R 5000 , N 4900

Figure 14: Figure 14: Boost factors WR



W_R 5000 , N 2500

Figure 15: Figure 15: Boost factors top

Delta R, Delta eta, Delta phi for each variables

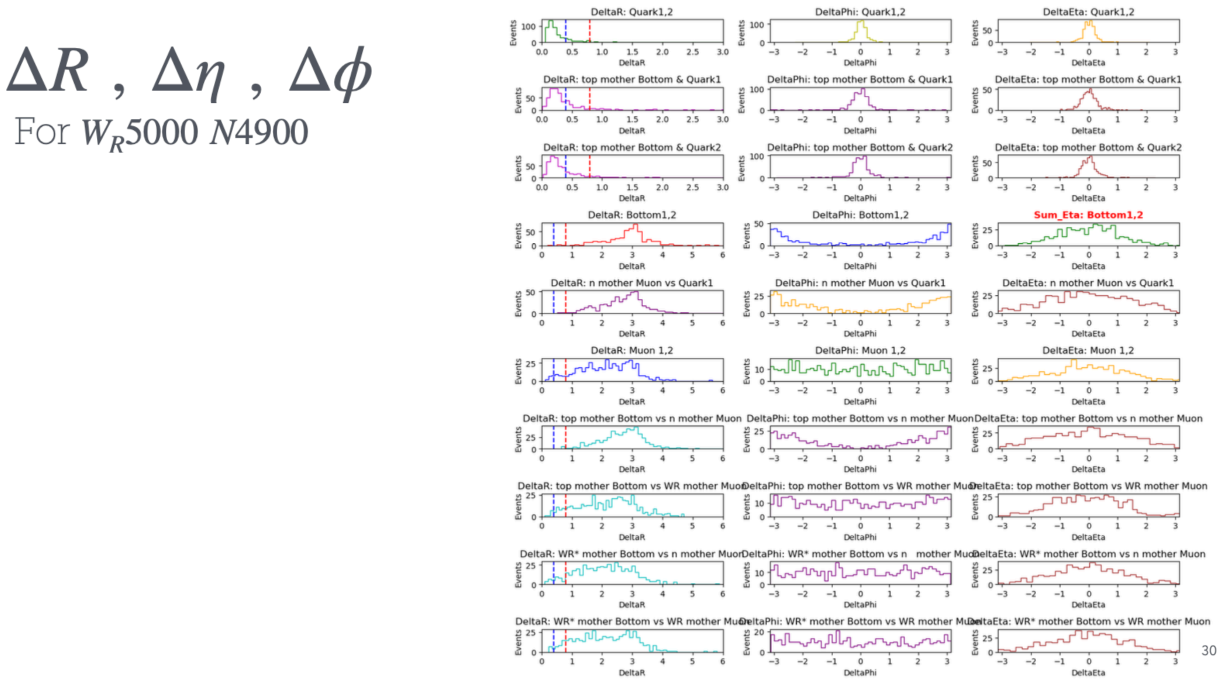


Figure 16: Figure 16: Angular correlations

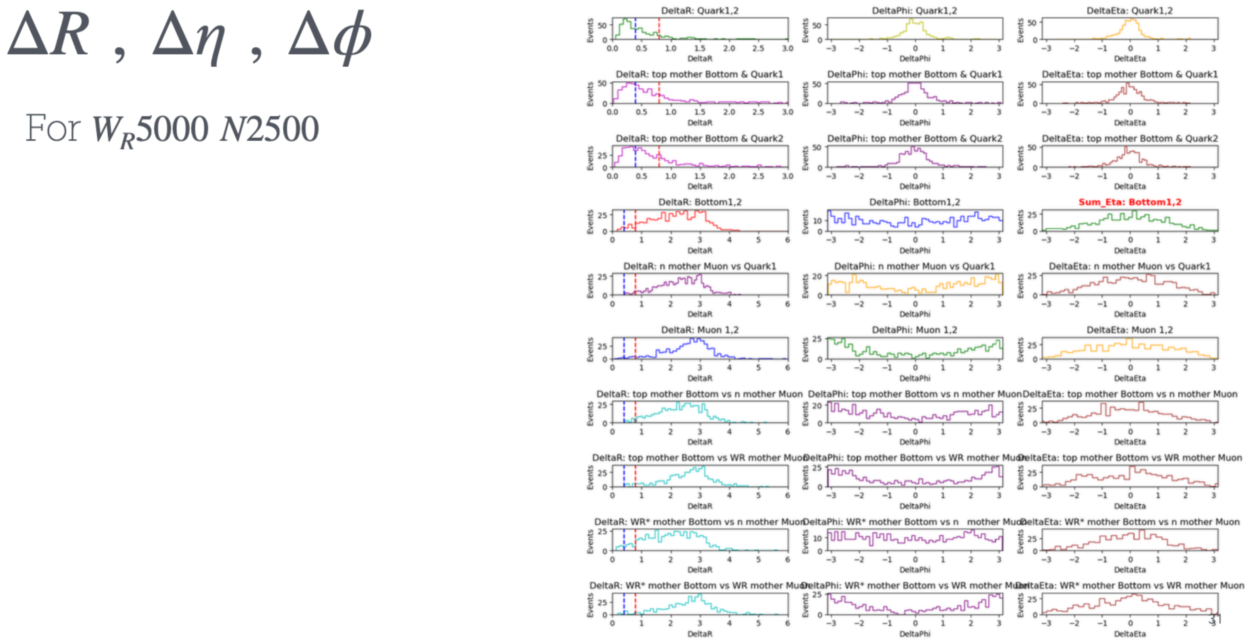


Figure 17: Figure 17: Additional angular distributions

2.1.1 WR ~ N Mass Region (5000, 4900 GeV) **Key observations**: - **WR* rest frame**: t and b quarks produced back-to-back (ΔR between bottom quarks ~3) - **Limited boost**: Since WR* not highly boosted, tb quarks separate significantly - **Lepton correlations**: - l1 (from WR): Can appear in all directions - l2 (from N): Back-to-back with WR* in N rest frame - **Unexpected behavior**: Top quark less boosted than expected, follows WR* direction - **Bottom quark**: Highly boosted, random direction relative to l2 - **Interpretation**: Suggests moderate WR* boost regime

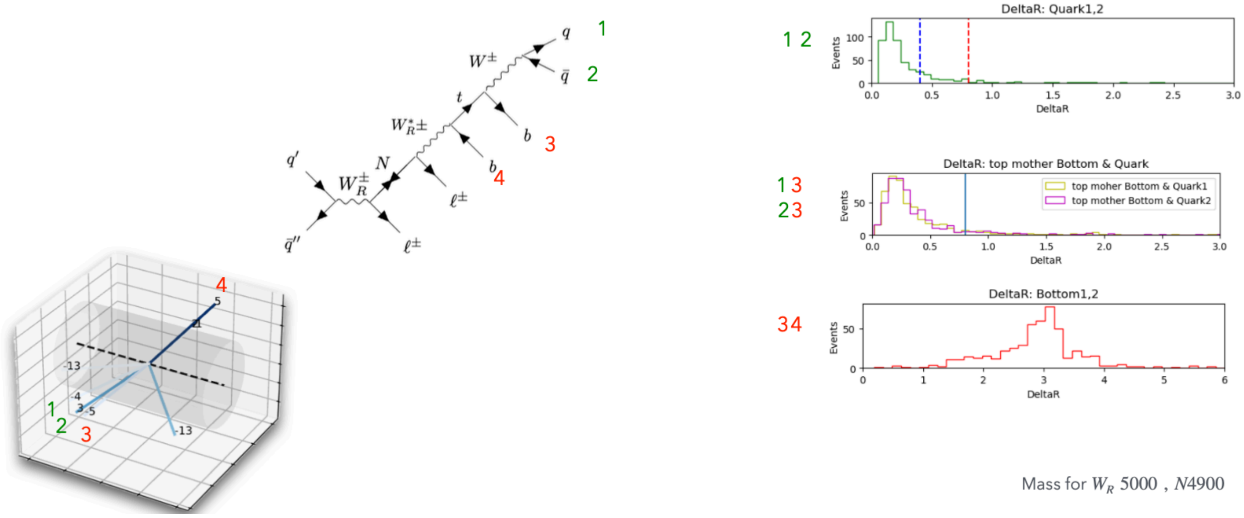


Figure 18: Figure 18: Kinematic analysis 4900 GeV case

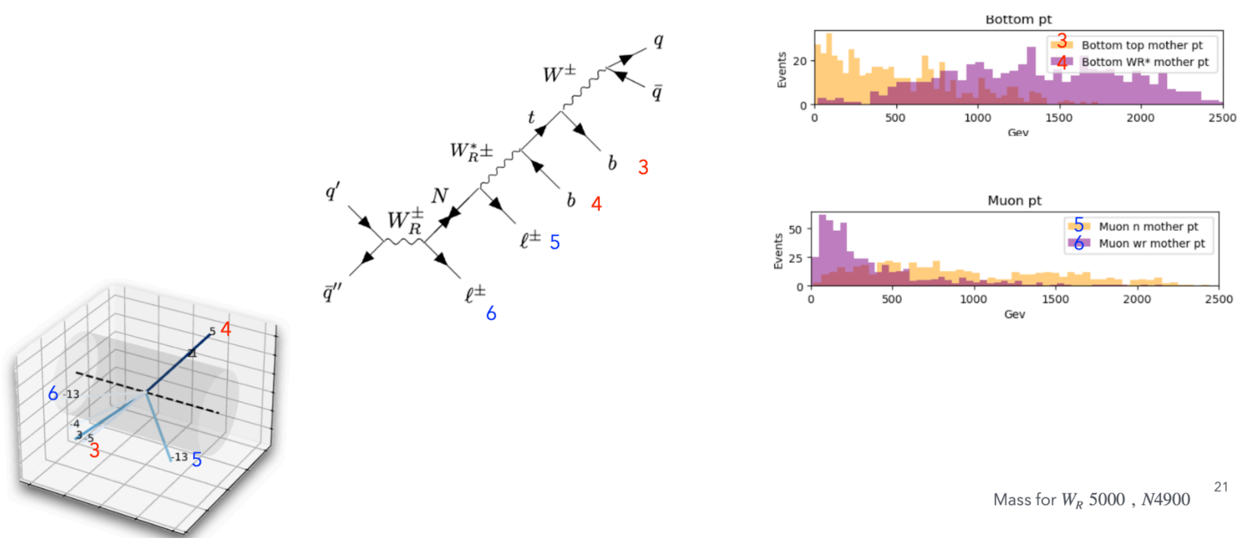


Figure 19: Figure 19: Additional kinematic plots

2.1.2 WR ~ N/2 Mass Region (5000, 2500 GeV) **Distinct topology observed**: - **Initial state**: N and l1 produced back-to-back, with N significantly boosted - **N decay dynamics**: - l2 partially opposes N boost → moderate displacement from WR* direction - Some l2 events still follow WR* direction due to boost competition - **WR* behavior**: Inherits N direction due to boost inheritance - **Final state tb decay**: - **b**

quark: Receives strong additional boost → significant separation from top jet - **Top jet**: Moderately boosted (reduced WR* mass) → partially follows N direction - **Angular correlation**: b quark direction depends on decay orientation: - Opposite WR* direction → back-to-back with system - Same WR* direction → collinear with top jet

2.2 AK8 Top Jet Analysis

Primary objective: Identify and characterize signal AK8 jets while understanding potential backgrounds and misidentification sources.

Analysis strategy: 1. **Signal jet identification**: Match closest AK8 jet to generator-level top quark 2. **Background characterization**: Study distance distributions and top scores for nearby AK8 jets 3. **Selection validation**: Test discriminating power of top tagger and soft-drop mass cuts 4. **Efficiency assessment**: Evaluate performance with randomized AK8 jet selection

2.2.1 Signal Efficiency and Fake Jet Studies **Objective**: Assess signal efficiency by identifying potential fake signals that mimic top jets.

Methodology: 1. **Geometric matching**: Examine ΔR between post-FSR generator-level top quarks and reconstructed AK8 jets 2. **Discriminant analysis**: Evaluate soft-drop mass and top tagger scores for nearby AK8 jets to assess misidentification probability

Results for WR = 5000, MN = 4900 GeV:

Key findings: - **Clean separation**: No overlap with other fat jets in signal region - **Low-mass artifacts**: Missing entries at low soft-drop mass due to poor jet reconstruction - **B-jet leakage signature**: Soft-drop mass range 80-150 GeV corresponds to top jets missing b-quarks - **Mass dependence**: WR~N/2 shows more b-quark leakage than WR~N due to reduced WR* boost → less collimated decay products

2.2.2 Signal Selection Efficiency **Overall performance assessment**: Application of top tagger and soft-drop mass cuts to complete signal sample

Efficiency results: 499/500 events (99.8%) successfully identify signal top quarks with AK8 jets

Mass point comparisons:

WR = 5000, MN = 4900 GeV:

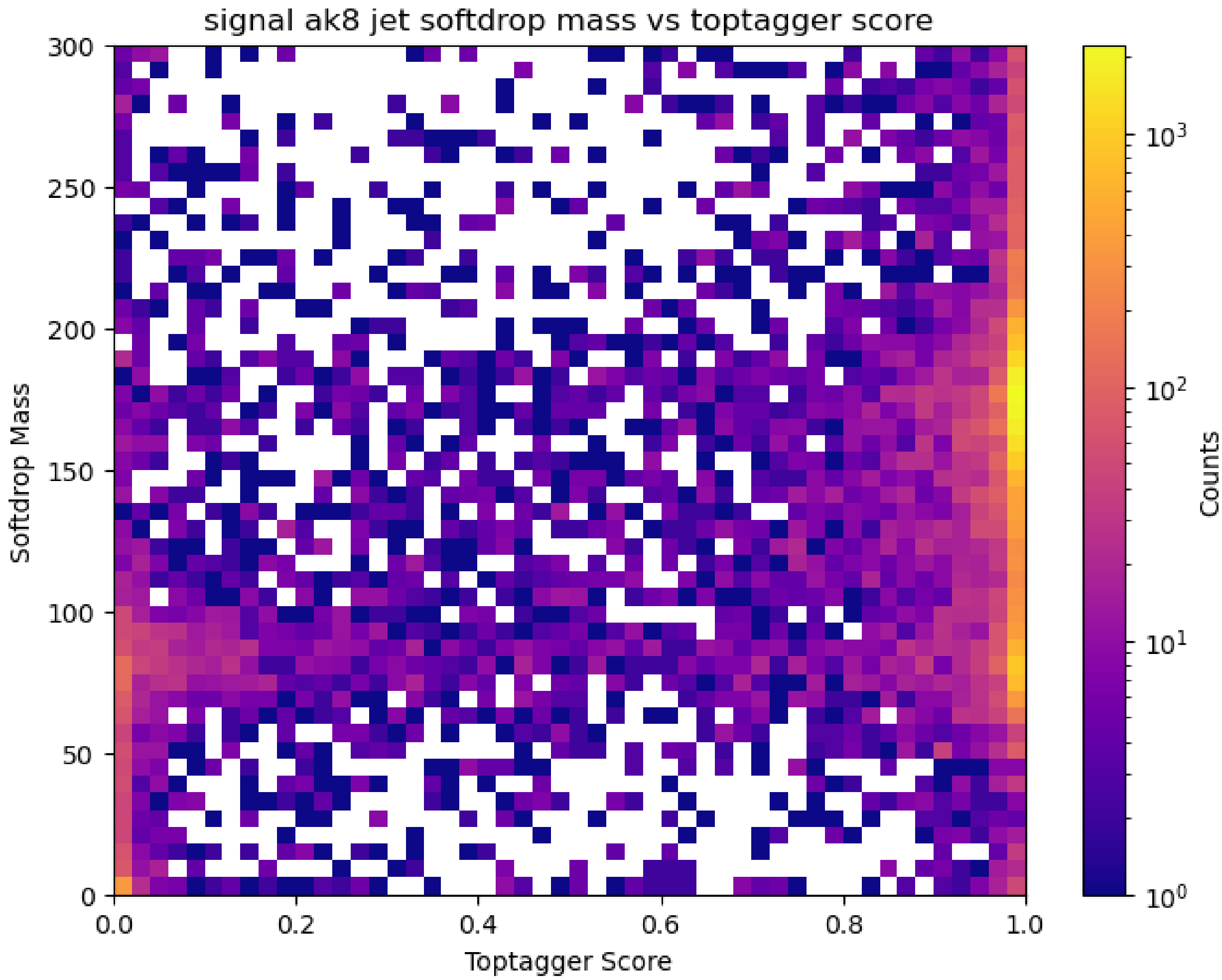


Figure 20: Figure 20: Top tagging efficiency - high N mass

2.2.3 Fake Jet Background Assessment **Random AK8 selection test**: Evaluate false positive rate when applying top tagging and soft-drop mass cuts to arbitrary AK8 jets

Key findings: - **Primary signal identification**: Closest ΔR match reliably identifies true signal - **Secondary contamination**: Some jets within ΔR < 0.8 pass selection cuts

2.2.4 Fake AK8 Jet Classification **Event categories**: No events with 3+ top-tagged jets observed

Classification scheme: 1. **Single AK8**: One jet passes selection 2. **Double AK8**: Two jets pass selection

Analysis for WR = 5000, MN = 4900 GeV **Double AK8 scenarios**: - **2S (two signal)**: Not possible → no additional AK8 jets within $\Delta R < 0.8$ of signal - **Event classification**: 1. **1F1S**: One jet within $\Delta R < 0.8$ + one jet outside $\Delta R > 0.8$ pass cuts 2. **2F**: Closest jet fails cuts + two jets outside $\Delta R > 0.8$ pass cuts

Event Type	2F (two fake)	1F1S (one fake + signal)
2 AK8 jets	Low rate	Moderate rate

Single AK8 scenarios:

Event Type	1F (one fake)	1S (signal only)
1 AK8 jet	Low rate	Dominant

Key observations: - **Single AK8 events**: No issues, dominant contribution with clear signal identification - **Double AK8 events**: Lower rate but challenging discrimination → top tagger shows limited separation, soft-drop mass distributions similar to genuine top jets

Angular Analysis and Interpretation **Hypothesis**: Double AK8 events may result from b-quark/top-quark angular correlation creating secondary jets outside $\Delta R > 0.8$

Investigation approach: 1. **LHE-level analysis**: Study η, φ separations between b quarks from different sources 2. **Reconstructed jet correlations**: Examine η, φ differences between AK8 jets and generator-level tops 3. **Mass correlation**: High soft-drop masses potentially due to WR*→b contamination 4. **Back-to-back topology**: Study large ΔR events with opposite φ values

Key questions: - How often do WR* b quarks fall outside $\Delta R = 0.8$ cone? - What fraction of single AK8 events contain hidden b quarks? - Are double AK8 events predominantly from b-quark leakage scenarios?

Conclusions from angular analysis: - **η difference/sum limitations**: Near-zero values make it difficult to distinguish back-to-back vs collinear configurations - **Dominant pattern**: Sum distribution peaks near 0, with very few events beyond 3 → confirms predominantly back-to-back topology - **Rare collinear cases**: Small fraction shows same-direction propagation - **Fake jet mechanism**: Collinear events likely create additional ΔR~0.8 cones from W-boson (from top) + b-quark (from WR*) combinations - **Event rate consistency**: No more than 2 AK8 jets pass baseline cuts, consistent with this interpretation

2.2.5 AK8 Jet Cleaning and Efficiency Losses For cleaning strategy, it must not overlap with signal muons that is selected and should not be overlapped with other ak8 jets. Did not consider overlapping with ak4 because it also can be reco with AK4.

After cleaning, top pt:

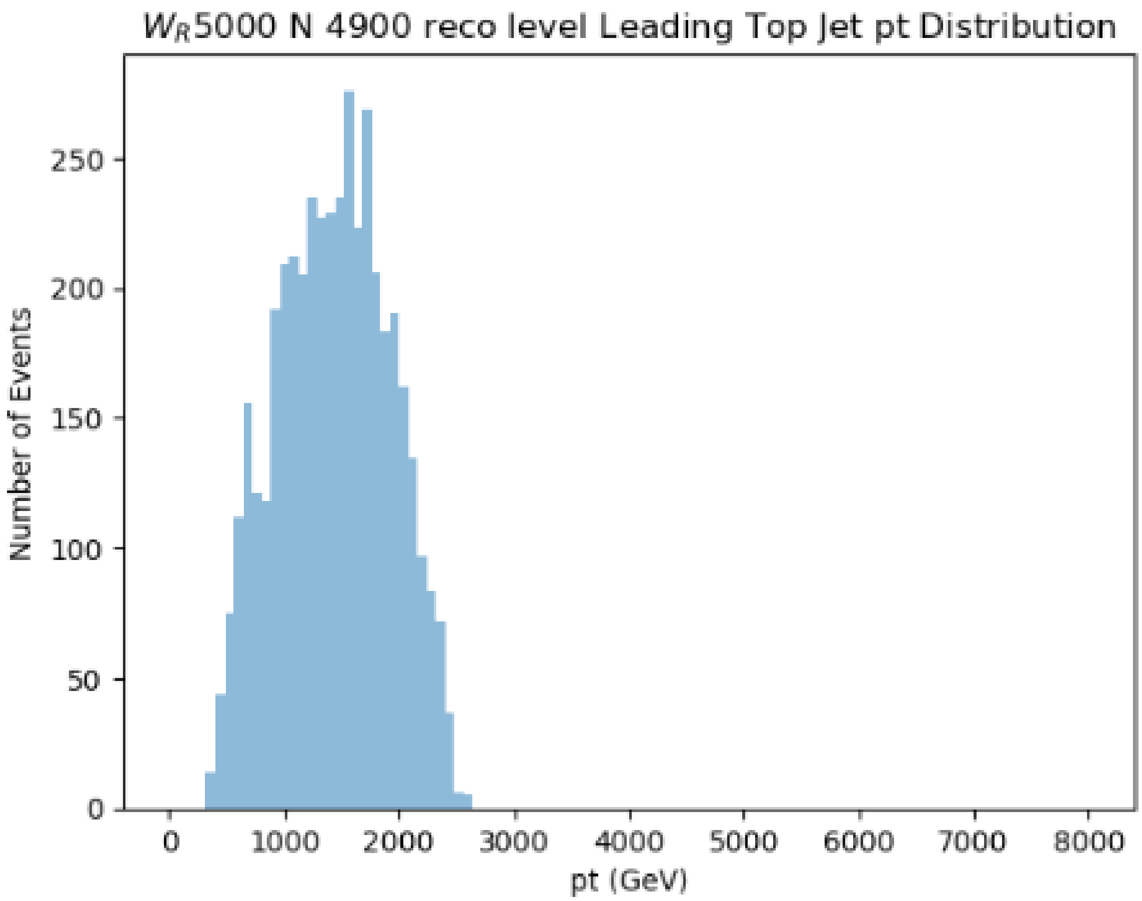


Figure 21: Figure 21: Top pT after cleaning

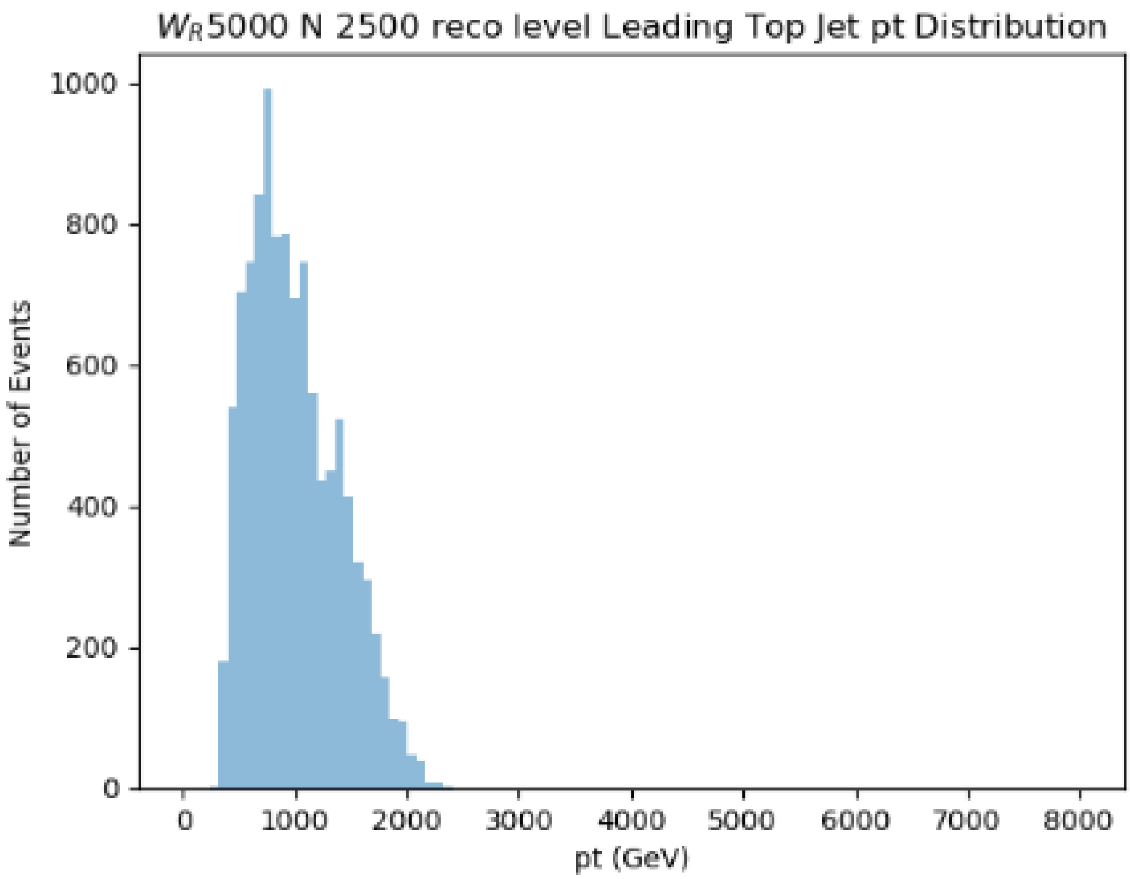


Figure 22: Figure 22: Additional cleaning results

2.2.5.1 B-Jet Cross-Contamination Analysis **Issue**: Back-to-back b-jets occasionally enter AK8 cones, though rate is relatively low

Method: Use LHE-level b-quarks to measure ΔR between signal b-jets and top jets for contamination assessment

Contamination rates:

WR = 5000, MN = 4900 GeV: - **WR* b-jets entering AK8**: 678/39,968 jets (1.2%) - **Top b-jets escaping AK8**: 3,303/39,968 jets (8.26%)

WR = 5000, MN = 2500 GeV: - **WR* b-jets entering AK8**: 4,710/56,759 jets (8.3%) - **Top b-jets escaping AK8**: 12,591/56,759 jets (22.18%)

2.2.6 Top tagging recommendation after cleaning

Reference: CMS Top Tagging Working Points

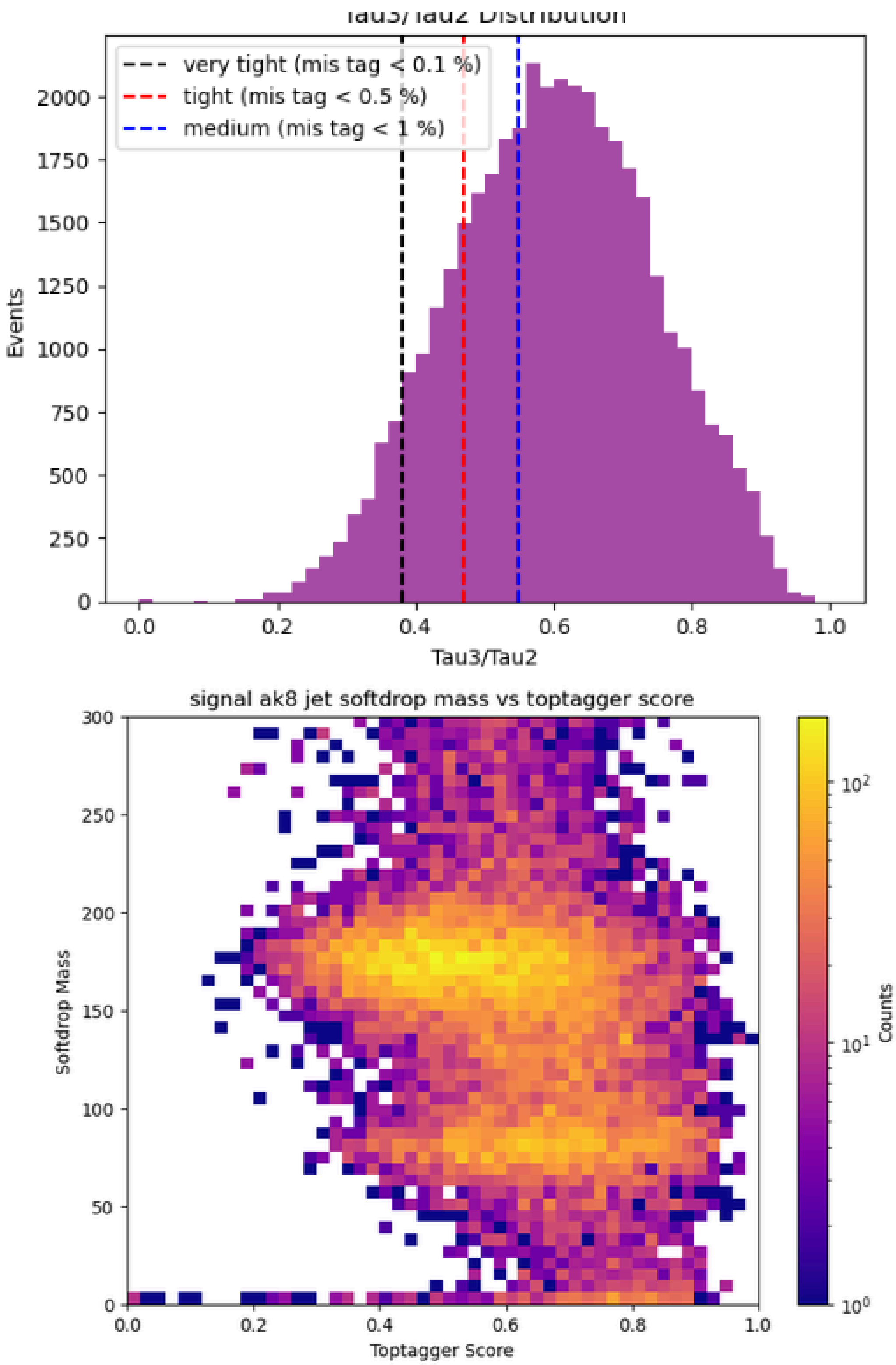


Figure 23: Figure 23: CMS Working Points

There is a WP with variable tau3/tau2 but it has low eff than particle net so It should be waitten until WP came with version NanoAOD v 15

For working process:

Figure 24: Figure 24: Working process 1

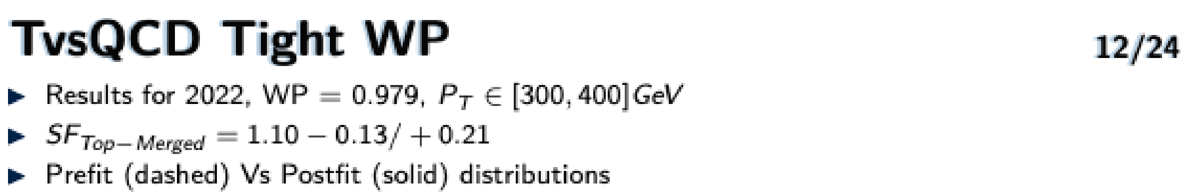


Figure 25: Figure 25: Working process 2

Top SFs

21/24

WP (Mist.Rate)	$300 \leq P_T \leq 400 GeV$	$400 \leq P_T \leq 480 GeV$	$480 \leq P_T \leq 600 GeV$	$600 \leq P_T \leq 1200 GeV$
0.98 (0.1%)	$1.10^{+0.21}_{-0.13}$	$0.83^{+0.08}_{-0.06}$	$0.86^{+0.07}_{-0.07}$	$1.04^{+0.12}_{-0.11}$
0.86 (0.5%)	$1.23^{+0.19}_{-0.15}$	$0.89^{+0.07}_{-0.06}$	$0.82^{+0.07}_{-0.07}$	$1.21^{+0.29}_{-0.18}$
0.68 (1%)	$1.22^{+0.20}_{-0.16}$	$0.89^{+0.06}_{-0.06}$	$0.84^{+0.07}_{-0.07}$	$1.27^{+0.29}_{-0.27}$

Table: 2022

WP (Mist.Rate)	$300 \leq P_T \leq 400 GeV$	$400 \leq P_T \leq 480 GeV$	$480 \leq P_T \leq 600 GeV$	$600 \leq P_T \leq 1200 GeV$
0.98 (0.1%)	$1.21^{+0.11}_{-0.10}$	$1.03^{+0.07}_{-0.07}$	$0.96^{+0.06}_{-0.06}$	$0.90^{+0.08}_{-0.07}$
0.87 (0.5%)	$1.38^{+0.15}_{-0.13}$	$1.04^{+0.51}_{-0.06}$	$1.01^{+0.06}_{-0.05}$	$0.89^{+0.08}_{-0.07}$
0.70 (1%)	$1.38^{+0.62}_{-0.12}$	$1.05^{+0.43}_{-0.05}$	$1.02^{+0.19}_{-0.05}$	$0.92^{+0.08}_{-0.07}$

Table: 2022EE

- High SFs for low P_T regions
- Stable (within error) as a function of P_T (for $P_T \geq 400$GeV)
- Asymmetric error for some 2022EE SFs

Figure 26: Figure 26: Working process 3

It might be better to use Loose WP due to high pt and for stable with error.

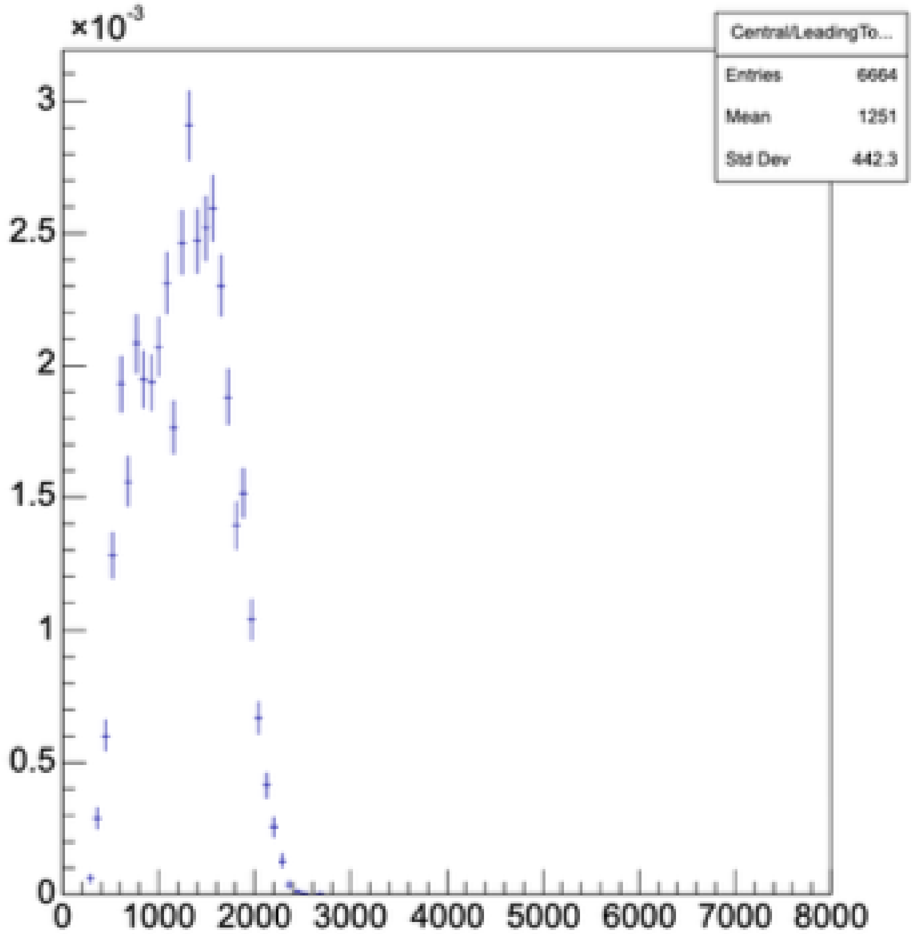


Figure 27: Figure 27: Cut adjustment

which this plot show when cut slight adjusted.

2.2.7 Conclusion **When two top jets are found**: If the masses are similar and they are adjacent to each other, there's no need to search for additional b-jets.

When only one top jet is present: - Case where b escapes from t and WR b doesn't enter: distinguishable by SDM - Case where b escapes from t and WR b enters: very rare, need to review if kinematically possible - Case where b doesn't escape from t and WR b enters: AK8 mass should be very large

Summary of events to discard:

AK8 Jet: 1. When top tagger 0.9, SDM 120 or higher are set as baseline (40% ~ 50%) → low soft drop mass cases, b escape cases 2. When two AK8s appear (when b jet protrudes forward) (5%) 3. When one AK8 appears - Among these, when WR*b enters (1% ~ 8%)

→ For WR 5000 N 4900 case, probability of obtaining clean and accurate top signal: ~55% → For WR 5000 N 2500 case, probability of obtaining clean and accurate top signal: ~40%

2.3 B-Jet Analysis

2.3.1 B-Jet Identification Efficiency **Objective**: Identify signal b-jets from WR* decay without relying primarily on jet mass reconstruction

Task: Identify AK4 jets originating from signal b-quarks

Geometric Matching Algorithm **Truth matching procedure**: 1. Use LHE b-quarks as ground truth for signal b identification 2. Find closest AK4 jet to each LHE b-quark using ΔR matching

Distance criteria observed: - **Primary matches**: Closest AK4 typically within ΔR < 0.4 (majority of cases) - **Secondary jets**: Second-closest jets predominantly fall outside ΔR > 0.4 - **Matching reliability**: LHE-to-closest-AK4 matching appears robust for signal identification

Performance Validation Studies **B-tagging algorithm comparison** (AK4 + AK8 + 2 leptons):

DeepJet algorithm [2022, Tight, 0.7183]: - Efficiency: 8.195% → 8.08% after HLT_IsoMu30 trigger

ParticleNet algorithm [2022, Tight, 0.6734]: - Efficiency: 8.63% → 8.465% after HLT_IsoMu30 trigger

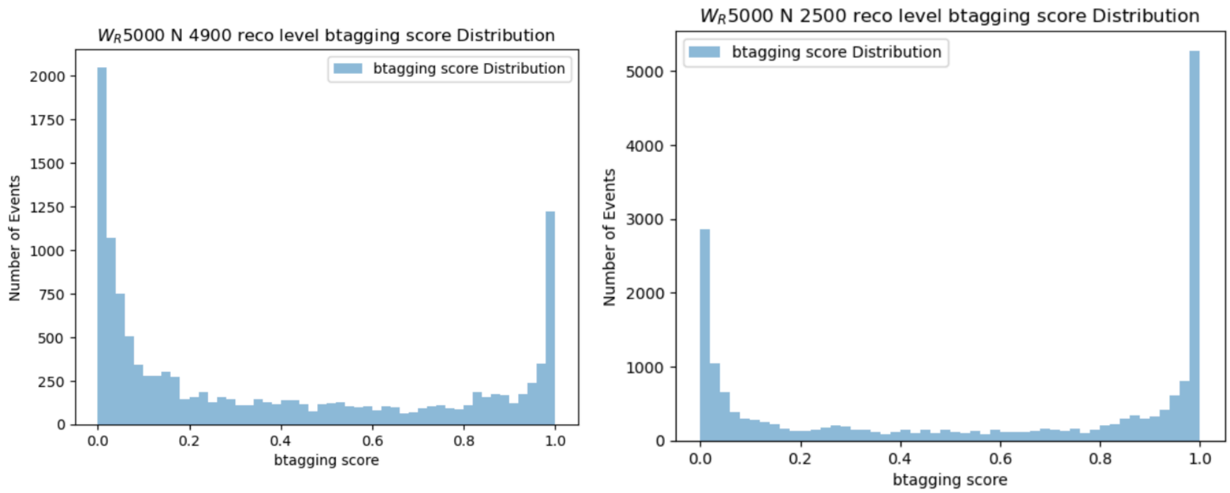


Figure 28: Figure 28: B-tagging performance

Combined selection efficiency (2 AK8 + 2 leptons): - Overall efficiency: 26.9% → 26.375% after HLT_IsoMu30 trigger

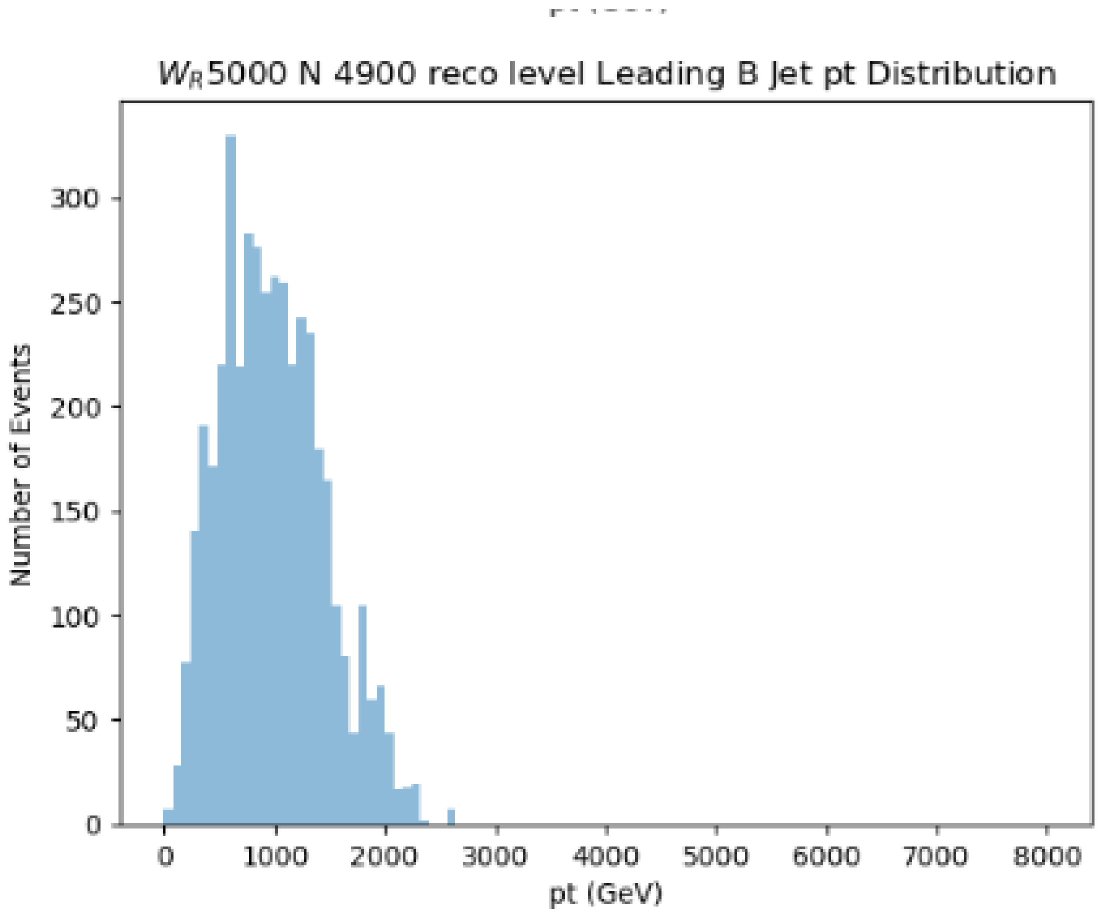


Figure 29: Figure 29: B-jet cleaning results

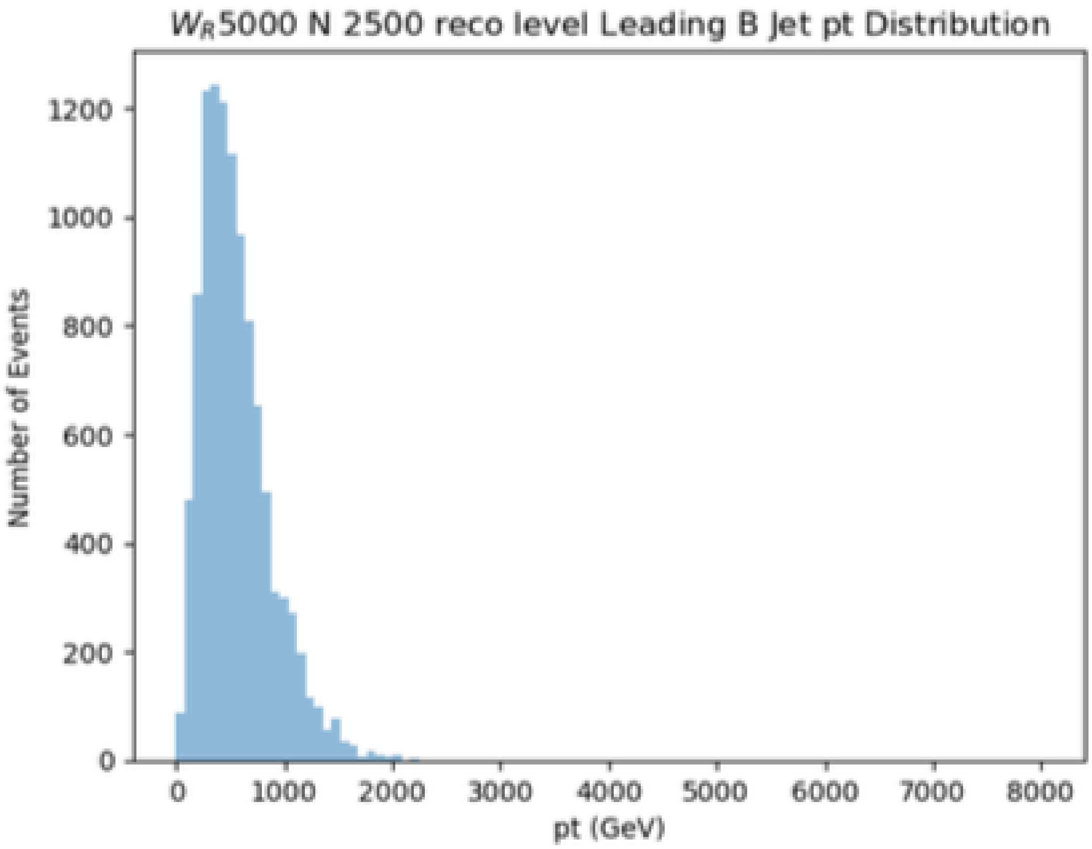


Figure 30: Figure 30: Additional b-jet results

2.3.2 After cleaning

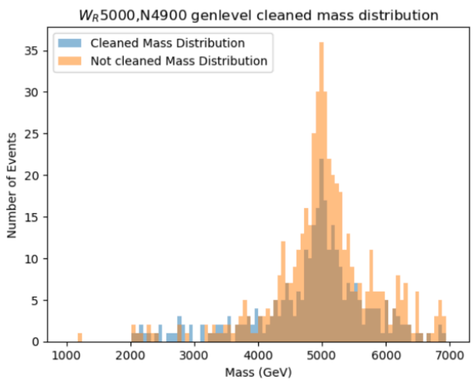
2.3.3 WR Mass Reconstruction Studies

Reconstruction scenarios tested for WR = 5000, MN = 2500 GeV:

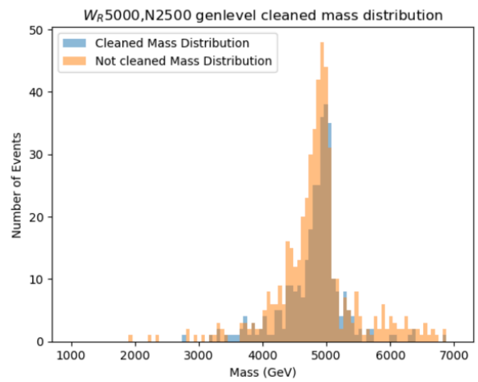
Same criteria selection in gen level

2 leptons , 2AK8

Not used top tagging , b tagging



83% efficiency



86% efficiency

Figure 31: Figure 31: Mass reconstruction comparison

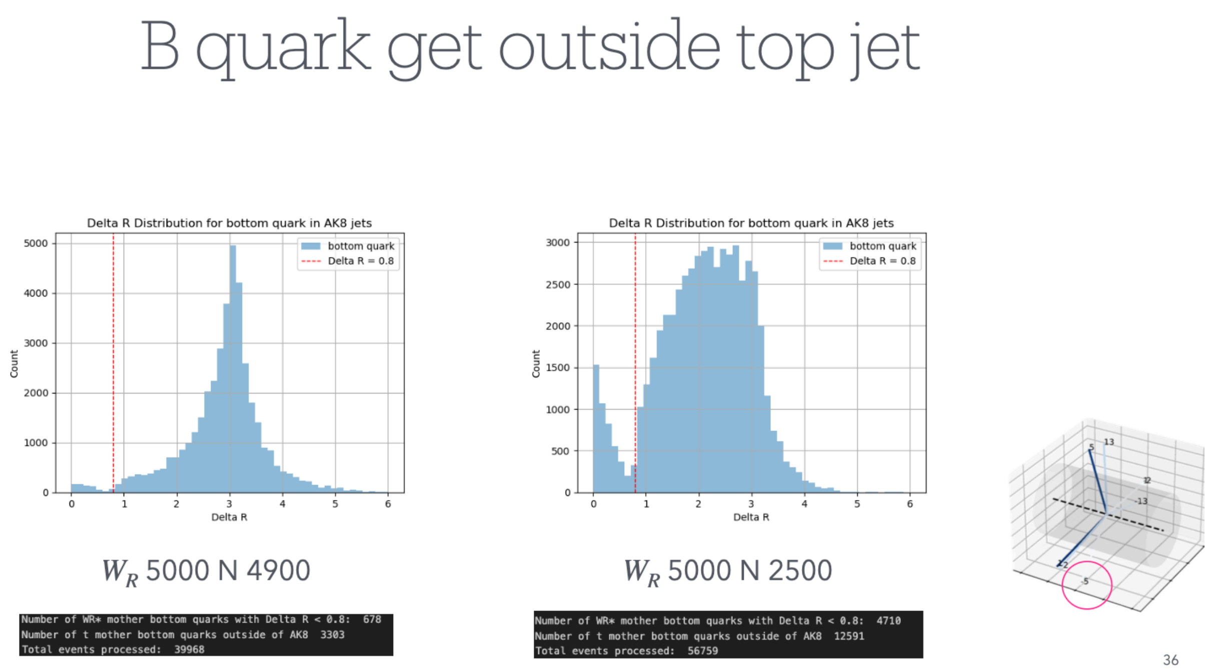


Figure 32: Figure 32: WR mass reconstruction

2.4 Lepton Analysis

Number of reco muon

```
Event 0 has 5 muons with pt: [551, 103, 47.3, 28.5, 27.7]
Event 1 has 7 muons with pt: [567, 370, 334, 51.9, 28.2, 21.7, 12.4]
Event 2 has 3 muons with pt: [924, 67.1, 15.5]
Event 3 has 10 muons with pt: [1.99e+03, 279, 253, 226, 98, 89.9, 57.3, 22.9, 13.2, 3.7]
Event 4 has 5 muons with pt: [519, 397, 80.2, 79.9, 27]
Event 5 has 4 muons with pt: [724, 391, 29.2, 16.8]
Event 6 has 5 muons with pt: [1.51e+03, 906, 35.6, 27.4, 24.5]
Event 7 has 7 muons with pt: [452, 219, 64.1, 31, 29, 17.3, 16.2]
Event 8 has 4 muons with pt: [557, 147, 18.7, 15.3]
Event 9 has 4 muons with pt: [1.11e+03, 83.4, 47.5, 3.5]
Event 10 has 4 muons with pt: [969, 121, 7.78, 5.04]
Event 11 has 4 muons with pt: [1.06e+03, 121, 29.6, 26.1]
Event 12 has 5 muons with pt: [534, 423, 21.2, 20.8, 4.64]
Event 13 has 3 muons with pt: [834, 258, 7.33]
Event 14 has 5 muons with pt: [545, 423, 13.6, 9.88, 5.85]
Event 15 has 5 muons with pt: [1.88e+03, 995, 162, 92.6, 15.6]
Event 16 has 8 muons with pt: [244, 80.8, 68.4, 66.7, 45.7, 38.9, 19.2, 6.26]
Event 17 has 7 muons with pt: [552, 154, 116, 28.5, 27.2, 25.4, 21.6]
Event 18 has 2 muons with pt: [158, 123]
Event 19 has 3 muons with pt: [1.48e+03, 874, 230]
Event 20 has 10 muons with pt: [301, 282, 193, 99.2, 75.8, 48.4, 23.5, 22, 20.4, 15.9]
Event 21 has 2 muons with pt: [462, 192]
Event 22 has 2 muons with pt: [2.4e+03, 190]
Event 23 has 2 muons with pt: [89.8, 15.8]
Event 24 has 9 muons with pt: [805, 621, 63.1, 52.9, 51.5, 48.6, 29.9, 11.3, 10.4]
...
Event 497 has 3 muons with pt: [1.93e+03, 197, 5.15]
Event 498 has 2 muons with pt: [959, 176]
Event 499 has 5 muons with pt: [632, 308, 114, 23.7, 22.2]
5.647294589178356 averge muon number per event
```

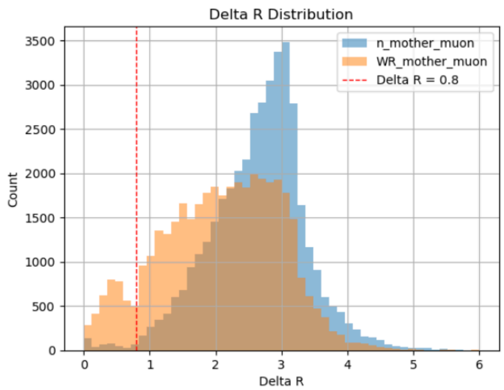
W_R 5000 , N 4900

```
Event 0 has 2 muons with pt: [1.19e+03, 779]
Event 1 has 3 muons with pt: [745, 648, 4.5]
Event 2 has 7 muons with pt: [1.58e+03, 1.02e+03, 328, 120, 114, 40.9, 8.53]
Event 3 has 6 muons with pt: [1.67e+03, 673, 40.6, 36.9, 31.8, 25.1]
Event 4 has 6 muons with pt: [1.07e+03, 513, 26.3, 25, 18.5, 13.9]
Event 5 has 4 muons with pt: [1.15e+03, 865, 137, 64.5]
Event 6 has 2 muons with pt: [863, 628]
Event 7 has 3 muons with pt: [1.67e+03, 628, 4.7]
Event 8 has 4 muons with pt: [1.13e+03, 881, 5.45, 5.3]
Event 9 has 3 muons with pt: [1.36e+03, 255, 9.25]
Event 10 has 4 muons with pt: [1.93e+03, 747, 5.08, 3.96]
Event 11 has 4 muons with pt: [1.25e+03, 1.09e+03, 89.7, 44.1]
Event 12 has 7 muons with pt: [1.78e+03, 314, 89, 49.2, 46.6, 34.4, 23.8]
Event 13 has 5 muons with pt: [1.71e+03, 1.53e+03, 609, 41.1, 32.6]
Event 14 has 4 muons with pt: [1.9e+03, 427, 28.5, 10.2]
Event 15 has 2 muons with pt: [1.63e+03, 1.16e+03]
Event 16 has 3 muons with pt: [1.19e+03, 268, 4.62]
Event 17 has 3 muons with pt: [1.42e+04, 1.43e+03, 602]
Event 18 has 4 muons with pt: [1.69e+03, 1.39e+03, 10.7, 7.6]
Event 19 has 4 muons with pt: [1.4e+03, 243, 53.2, 6.67]
Event 20 has 3 muons with pt: [1.75e+03, 373, 21.8]
Event 21 has 4 muons with pt: [1.67e+03, 1.45e+03, 8.4, 4.85]
Event 22 has 2 muons with pt: [1.48e+03, 1.02e+03]
Event 23 has 3 muons with pt: [971, 650, 152]
Event 24 has 4 muons with pt: [1.15e+03, 476, 106, 69.1]
...
Event 497 has 4 muons with pt: [743, 666, 366, 3.27]
Event 498 has 7 muons with pt: [694, 243, 38.1, 27.2, 17.1, 13.8, 3.72]
Event 499 has 5 muons with pt: [1.04e+03, 944, 18.8, 15.8, 4.04]
4.040080160320641 averge muon number per event
```

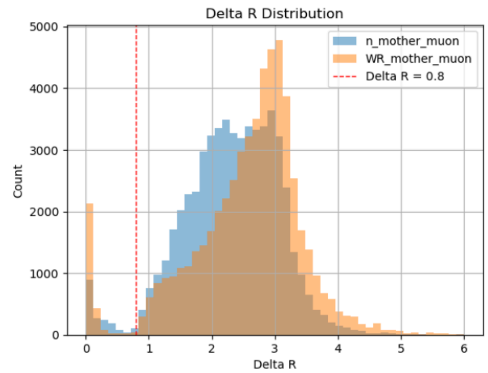
W_R 5000 , N 2500

Figure 33: Figure 33: Muon features

Number of muon inside top jets



W_R 5000 , N 4900



W_R 5000 , N 2500

Gets inside ~ 10%

Figure 34: Figure 34: Additional muon properties

Muon p_t distribution with signal & bkg

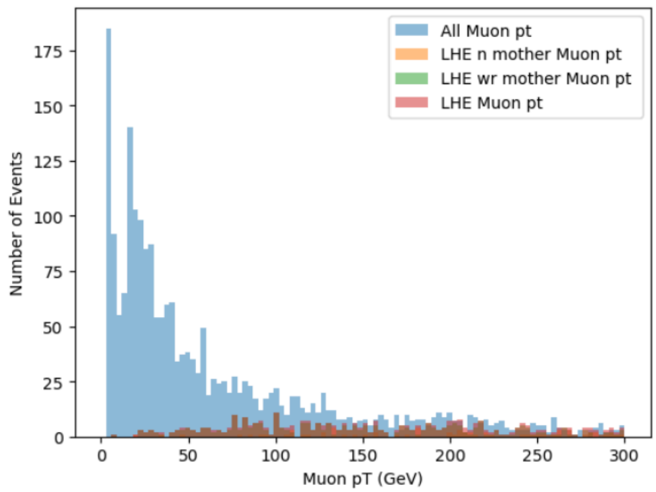


Figure 35: Figure 35: Muon distributions

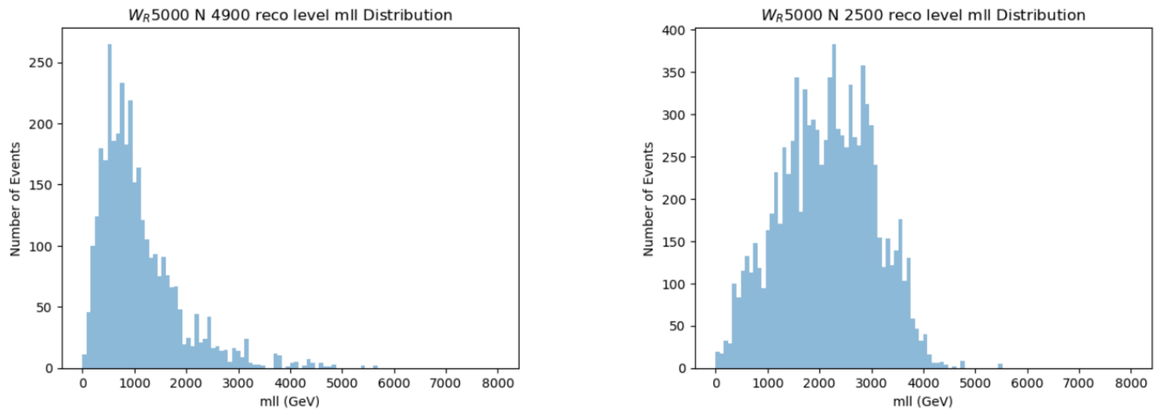


Figure 36: Figure 36: Muon analysis

2.4.1 Muon features

2.4.2 Lepton cleaning **Cases where they entered this ak8** (b jet overlap cases were not many in 5000 4900): → It seems we can use leptons from lhe and use the found fatjet.

WR 5000 N2500: survived jet: 52313 // cleaned jet: 4410 // total jet: 56723 @ On average, 4 muons are contained.

WR 5000 N 4900: survived jet: 36003 // cleaned jet: 3953 // total jet: 39956 @ On average, 6.5 muons are contained.

2.4.3 Lepton ID, Trigger selection To get Leptons, first it should go through HLT, and we should choose leptons with ID.

For Muon, checking features first:

1. pt

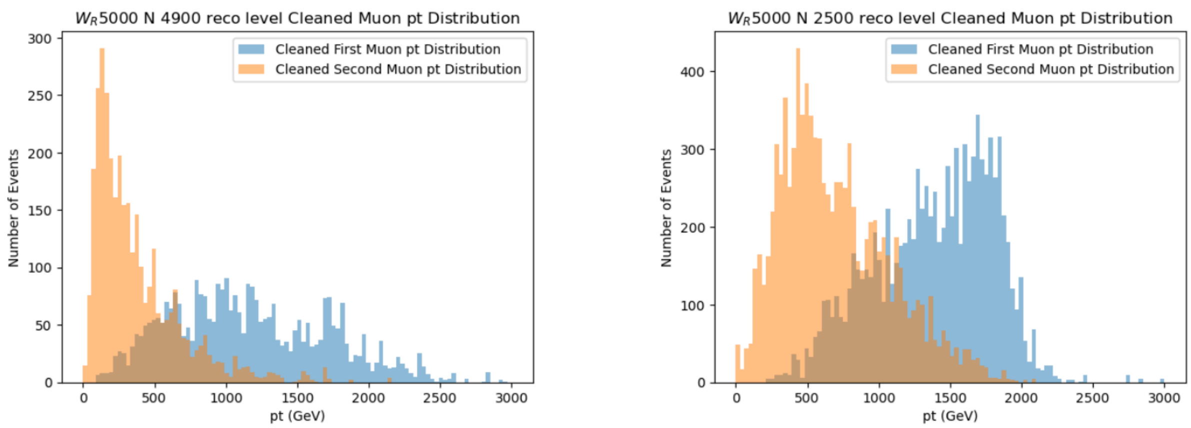
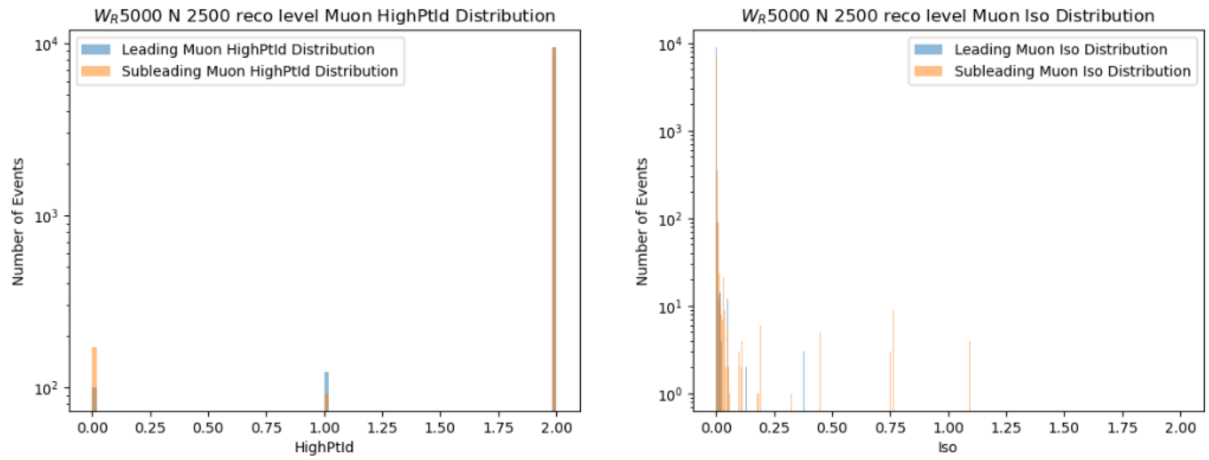


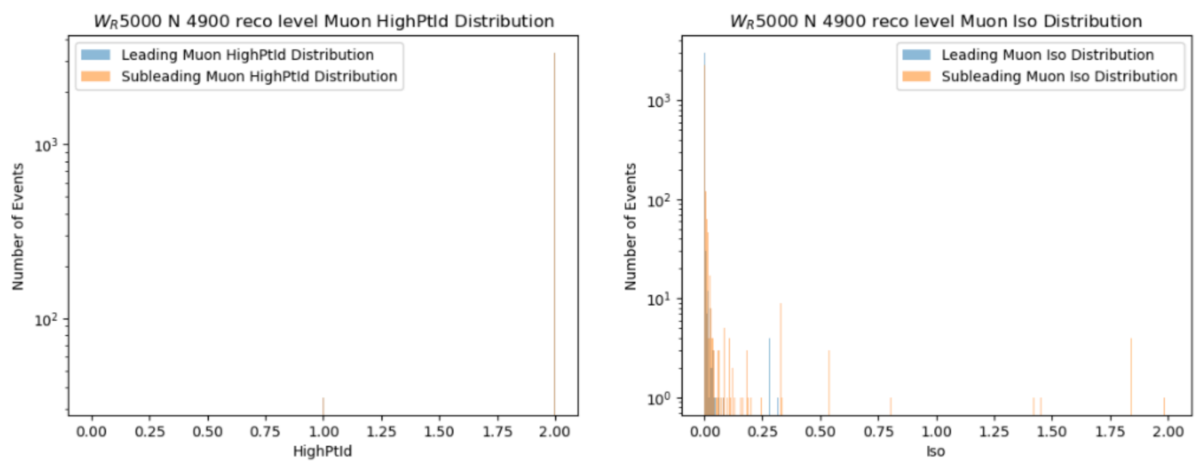
Figure 37: Figure 37: Muon pT

2. High pt id and isolated



[Muon_highPtId] [Muon_tkRelIso]

Figure 38: Figure 38: High pT ID



[Muon_highPtId] [Muon_tkRelIso]
(1 : tracker high p_t ID
2: global high p_t ID)

Figure 39: Figure 39: Isolation

which mean we can use high pt id, and iso id, must also check recommendation.

HLT recommendation gives:

Event type	Recommended path	Comments
Single muon, intermediate pT	HLT_IsoMu24_v*	- Unprescaled for the whole year - final filter: hltL3crIsoL1sSingleMu22L1f0L2f10QL3f24QL3trkIsoFiltered
Single muon, high pT	HLT_Mu50_v* OR HLT_CascadeMu100_v* OR HLT_HighPtTkMu100_v*	- Unprescaled for the whole year - final filters: hltL3fL1sMu22Or25L1f0L2f10QL3Filtered50Q, hltL3fL1sMu22Or25L1f0L2f10QL3Filtered100Q, hltL3fL1sMu25f0TkFiltered100Q - OR with CascadeMu100 & HighPtTkMu100: recover the inefficiency on high pT (~O(100)GeV) muons due to global chi2/ndof cut in Mu50 trigger (details: slides)
Double muon, intermediate pT	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v*	- Unprescaled for the whole year

Figure 40: Figure 40: HLT recommendations

And for ID's:

Particle-Flow (PF) Algorithm

- **전체 이벤트의 글로벌 정보**를 활용하여 모든 입자(charged hadrons, neutral hadrons, photons, electrons, muons)를 종합적으로 재구성
- 각 입자의 에너지와 운동량을 **모든 서브검출기 정보를 조합**하여 최적화
- **이벤트 레벨에서의 일관성**을 보장

Non-PF (Traditional) Approach

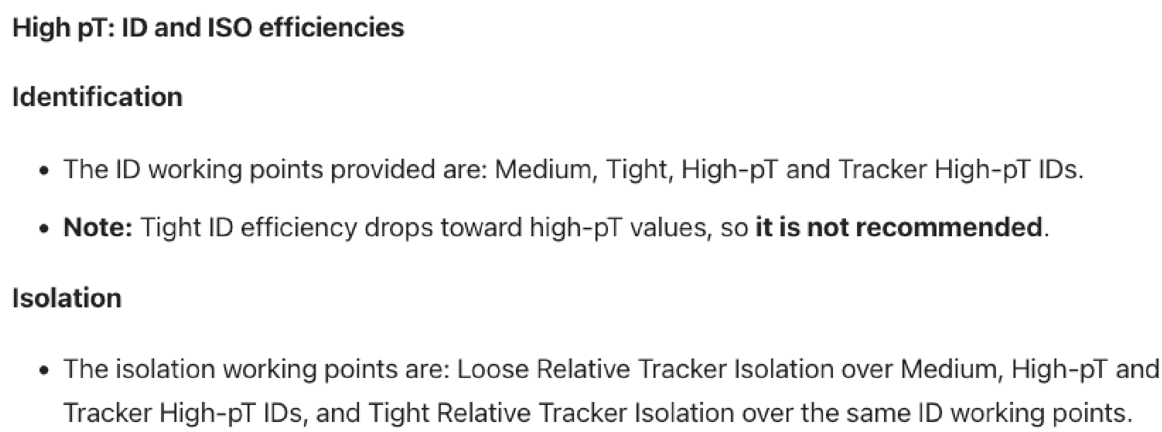
- **뮤온 검출기와 트래커 정보만**을 사용하여 뮤온을 재구성
- 다른 입자들과의 상호작용을 고려하지 않음
- **독립적인 뮤온 재구성**에 집중

2. 주요 차이점

특징	PF-based (Tight ID)	Non-PF-based (High-pT ID)
적용 범위	pT < 200 GeV 권장	**pT > 200 GeV 전용**
알고리즘	Particle-Flow 종합 재구성	TuneP 알고리즘 사용
정보 활용	전체 이벤트 정보	뮤온 검출기 + 트래커만
효율성	pT > 200 GeV에서 **효율성 저하**	고에너지에서 **효율성 유지**
가짜 비율	낮은 fake rate	상대적으로 높은 fake rate
운동량 할당	PF 기반 최적화	**TuneP 기반 최적화**

Figure 41: Figure 41: Muon ID recommendations

Conclusion: For muon selection, use: - **HLT**: HLT_Mu50 || HLT_HighPtTkMu100 || HLT_CascadeMu100 due to pt over 1 TeV - **ID**: HighptID == 2 (tight), Muon_tkIsoId == 2 (tight) - **pT cut**: starts from 50 GeV for SF availability


High pT: ID and ISO efficiencies

Identification

- The ID working points provided are: Medium, Tight, High-pT and Tracker High-pT IDs.
- **Note:** Tight ID efficiency drops toward high-pT values, so **it is not recommended**.

Isolation

- The isolation working points are: Loose Relative Tracker Isolation over Medium, High-pT and Tracker High-pT IDs, and Tight Relative Tracker Isolation over the same ID working points.

Figure 42: Figure 42: Final muon selection

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3. Background Analysis

Selection cuts comparison:

Cut	HLT	lead Muon pt	sub lead muon pt	Bjet	B pt	Top score	Top SDM	Top pt	WR mass	m(ll)
cut1	ISOMU30	>30	0	Tight	30	0.9	120 250	0	2000	300
cut2	ISOMU30	>30	0	Tight	300	0.9	120 250	500	2000	300

cut1: BKG → TTHtononbb, TTLJ, TTTT, TTWtonu, TTZ, TZq

cut2: BKG → ST_TW_antitop_Semilep, ST_tW_top_Semilep, TTG_PTG200toinf, TTHToNonbb, TTLJ, TTTT, TTWtoNu, TTZ_M4to50, TTZ_M50, TTZ_NoFullyHad

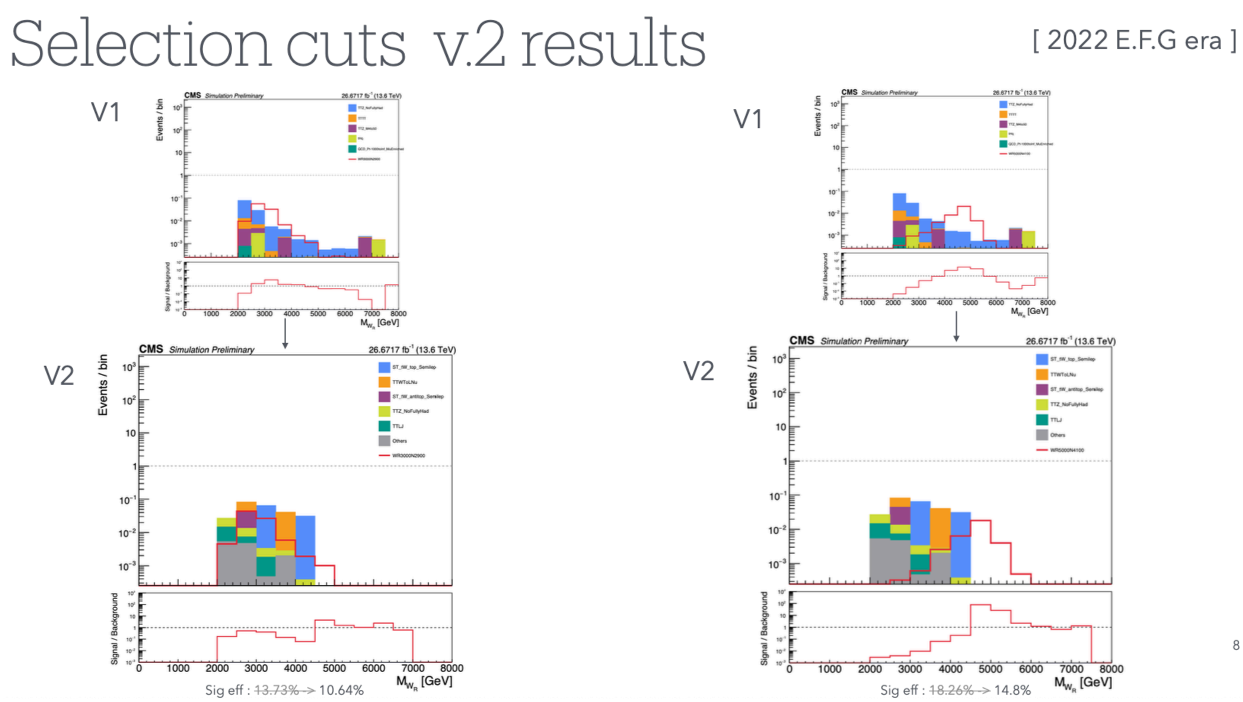

Figure 43: Figure 43: Background analysis

Issues with these cuts:

1. Muon trigger is not a recommended one, also it is isolated and high pt, which can use other HLT
2. No Muon ID selected, also pt cut must be chosen with Muon ID min pt
3. For cut2 B pt is too high to adjust scale factors. It should have some pt to measure scale factor which we can observe in data
4. Also mll cut or wr mass cut gives large constraints of background, which makes it not available to measure SF which is the discrepancy of data and signal

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Summary and Conclusions

This analysis presents a comprehensive study of the LRSM WR→tb channel, focusing on the WR ~ N mass region to optimize signal reconstruction and background rejection. Key findings include:

1. **Kinematic Understanding**: The WR ~ N mass region provides better separation of decay products compared to the highly boosted regime

2. **Top Jet Reconstruction**: AK8 jets with ParticleNet top tagging achieve ~55-40% efficiency after cleaning
3. **B-jet Identification**: AK4 b-jets can be identified with moderate efficiency using geometric matching and flavor tagging
4. **Lepton Selection**: High-pT muon triggers and tight ID requirements are essential for signal selection
5. **Background Challenges**: Careful optimization of selection cuts is needed to maintain sensitivity while allowing for scale factor measurements

Future work should focus on optimizing the selection criteria and developing data-driven background estimation methods.

Appendix A: Technical Details

A.1 Software and Analysis Framework

Analysis Tools: - MadGraph for signal generation - ROOT framework for data analysis - NanoAOD format for event storage - ParticleNet for top/b-tagging

A.2 Selection Criteria Summary

Final Recommended Cuts:

Object	Selection Criteria	Efficiency
Muons	pT > 50 GeV, HighPtID=2, tkIsoId=2	~90%
AK8 Jets	pT > 200 GeV, ParticleNet > 0.9, 120 < SDM < 250 GeV	~55-40%
AK4 b-jets	pT > 30 GeV, DeepJet Tight WP	~8%
HLT	HLT_Mu50 OR HLT_HighPtTkMu100 OR HLT_CascadeMu100	~95%

A.3 Systematic Uncertainties

Expected Sources: - Jet Energy Scale/Resolution - b-tagging scale factors - Top-tagging scale factors - Muon ID/trigger scale factors - Luminosity uncertainty - PDF uncertainties

A.4 Background Composition

Dominant Backgrounds: - $t\bar{t}$ + jets (dominant) - Single top - $t\bar{t}$ + V (V = W, Z) - $t\bar{t}$ + H - Diboson production

A.5 Signal Optimization Strategy

Mass-dependent Approach: - WR ~ N region: Resolved analysis with separated objects - High WR region: Boosted analysis with merged jets - Transition region: Hybrid approach

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References

1. CMS Collaboration, "Search for heavy right-handed W bosons in events with an electron and missing transverse momentum"(EXO-20-002)
2. CMS Top Tagging Working Group recommendations
3. CMS B-tagging Performance Studies, Summer22
4. CMS Muon Performance Studies, Run 3
5. Left-Right Symmetric Model theoretical framework

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